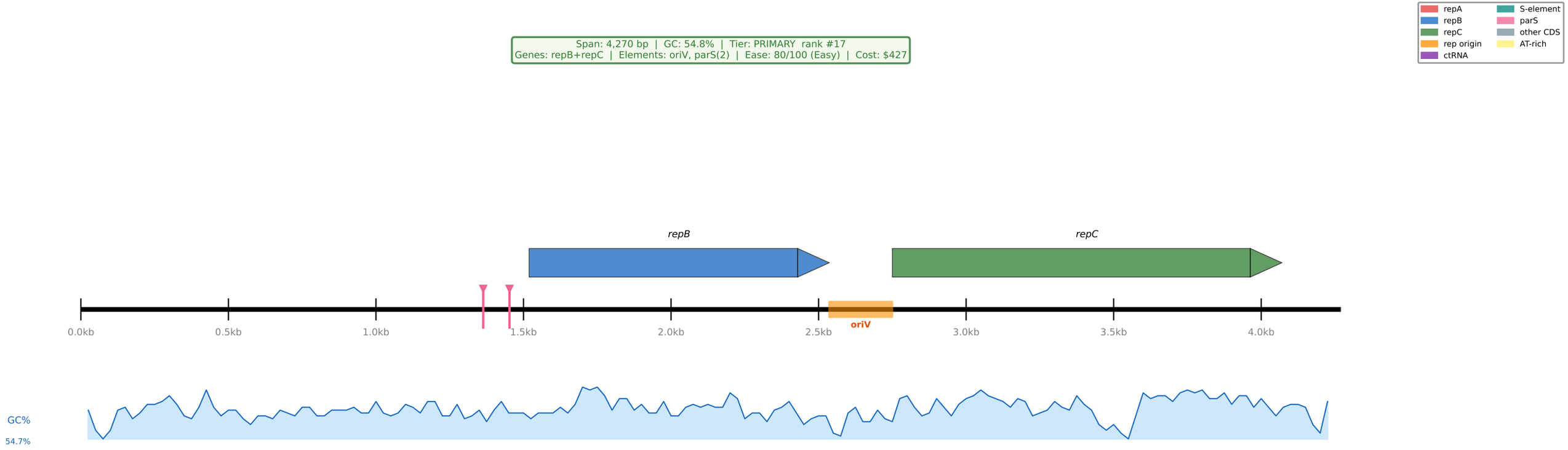
Final Synthesis / Cloning Order

73 origins | 244,813 bp of origin + 4,161 bp of barcode cassette | 248,974 bp total

```
Origin bp (plotted on the following pages):  244,813
Cassette bp (57 bp per origin, NOT shown):     4,161
Total order bp:                              248,974

Pool breakdown:
  repABC_primary                 26 origins
  repABC_alt_cloning              7 origins
  diversity_PLSDB                40 origins

Production method:
  twist_synthesis                66 origins
  genomic_pcr_amplification       7 origins

Diversity-tier mix:
  Tier 3: Cross-phylum BHR            5
  Tier 2: Alphaproteobacteria        29
  Tier 4: Phylum-exclusive            6

Barcode cassette (top-strand 5'→3', appended to 3' end of each origin):
  A  |  GATGTCCACGAGGTCTCT  |  [N20]  |  CGTACGCTGCAGGTCGAC
       (spacer)        (U1)         (barcode)        (U2)

Barcodes are in final_order.csv (barcode_N20, barcoded_sequence) and
final_order_barcoded.fasta. See barcode_diversity_report.{txt,pdf}
for the Hamming/Levenshtein/positional QC audit.

The maps on the following pages are the structural annotations from:
  - analyze_repABC_origins.py  (repABC, from ARC_repABC_loci_fna_gbk)
  - plasmid_origin_pipeline.py (diversity, with Pfam-annotated ORFs)
```

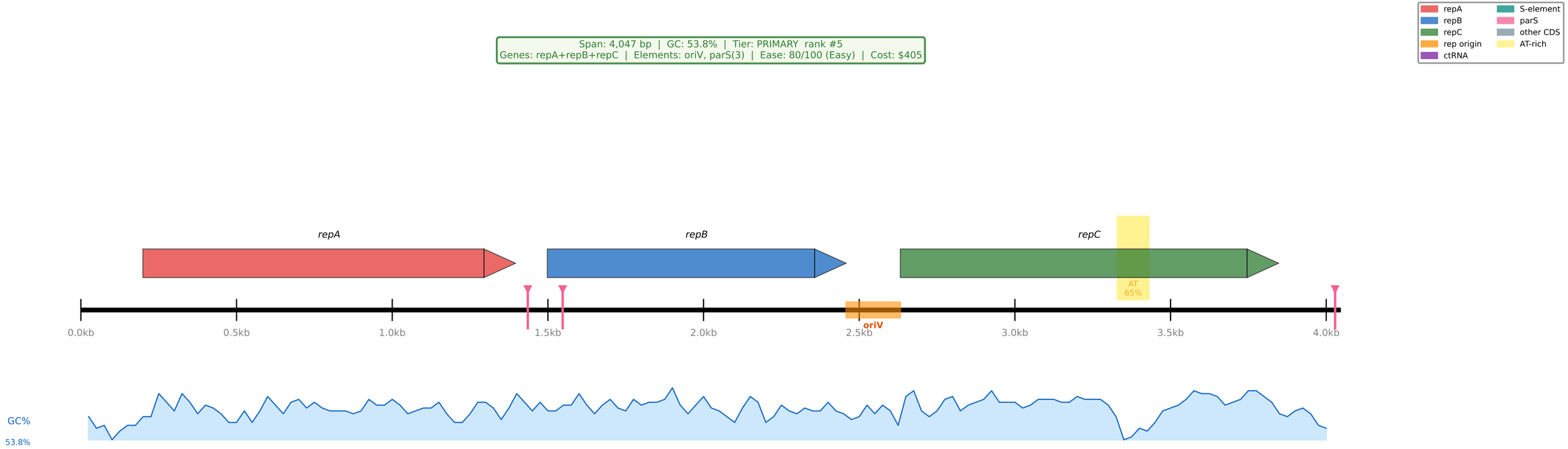
15-1187-1-2a_CP123842.1_pTi_Type_III_rep_1 | Type III rep 1 | Agrobacterium tumefaciens

Span: 4,270 bp | GC: 54.8% | Tier: PRIMARY rank #17
Genes: repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $427



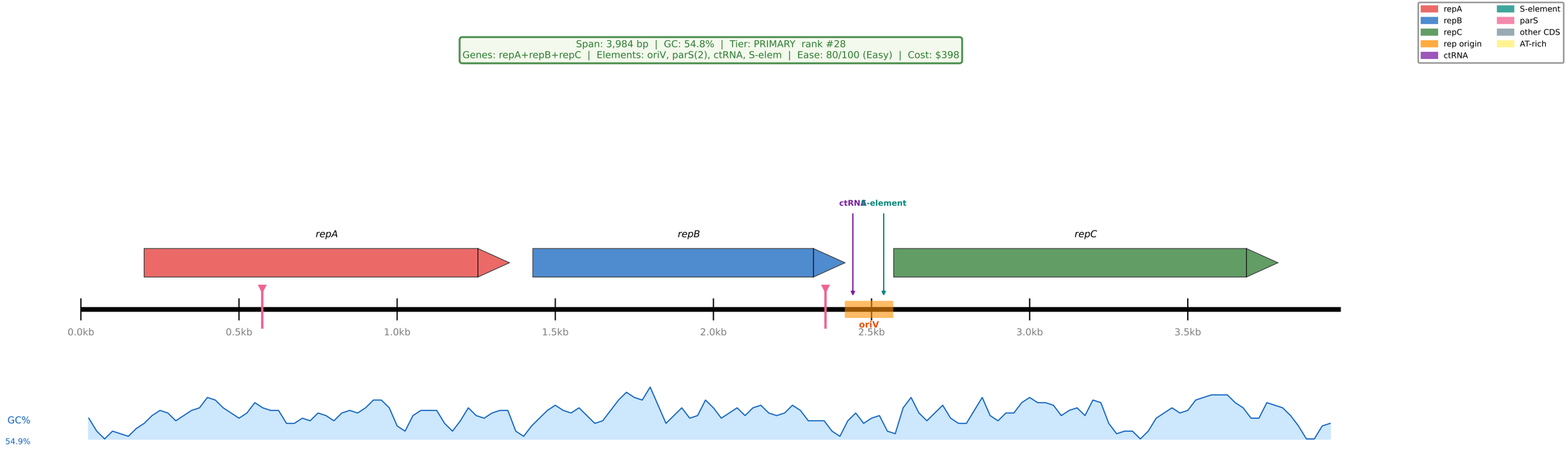
15-1187-1-2a_CP123842.1_pTi_Type_III_rep_2 | Type III rep 2 | Agrobacterium tumefaciens

Span: 4,047 bp | GC: 53.8% | Tier: PRIMARY rank #5
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $405



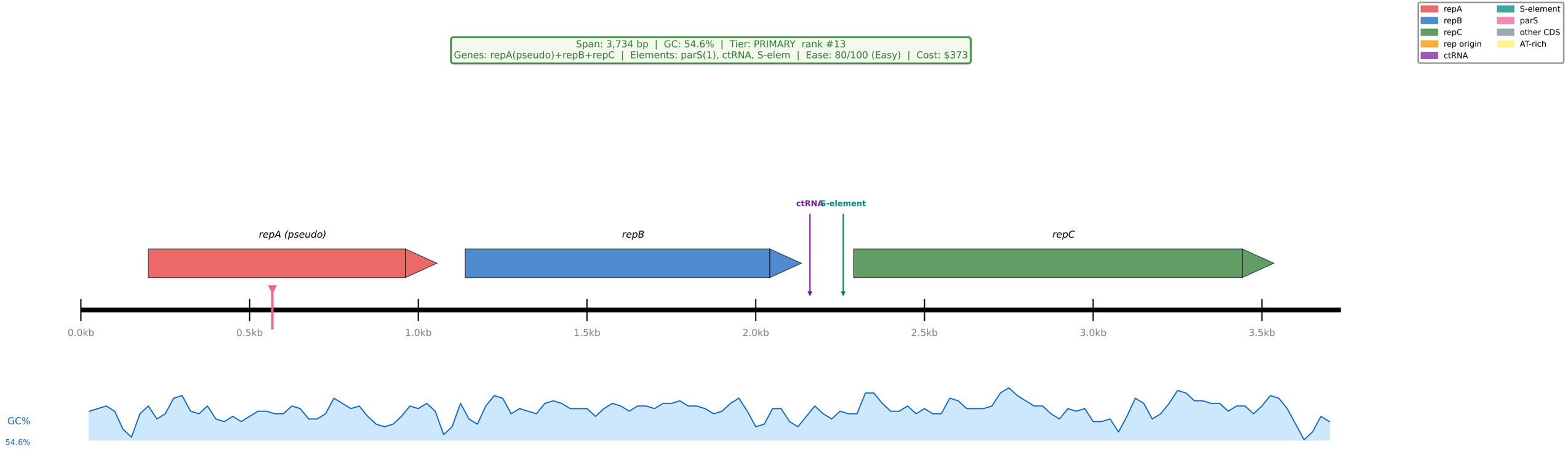
1D1514_plasmid1 | unknown | Agrobacterium tumefaciens

Span: 3,984 bp | GC: 54.8% | Tier: PRIMARY rank #28
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



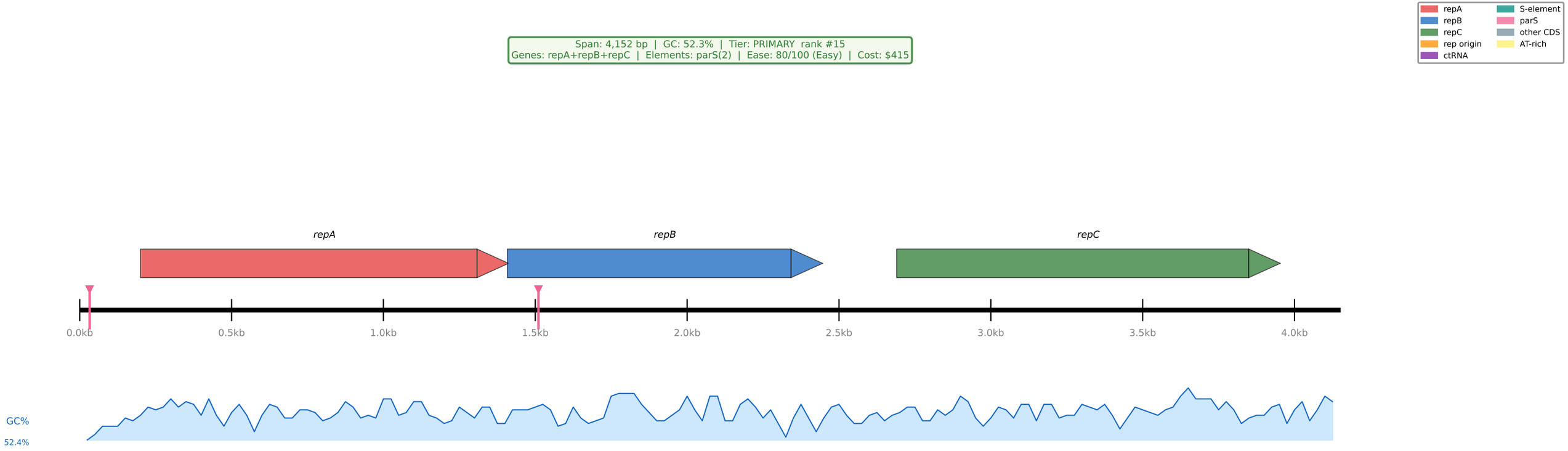
1D1514_plasmid2_pTi_Type_V_rep_1 | Type V rep 1 | Agrobacterium tumefaciens

Span: 3,734 bp | GC: 54.6% | Tier: PRIMARY rank #13
Genes: repA(pseudo)+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $373



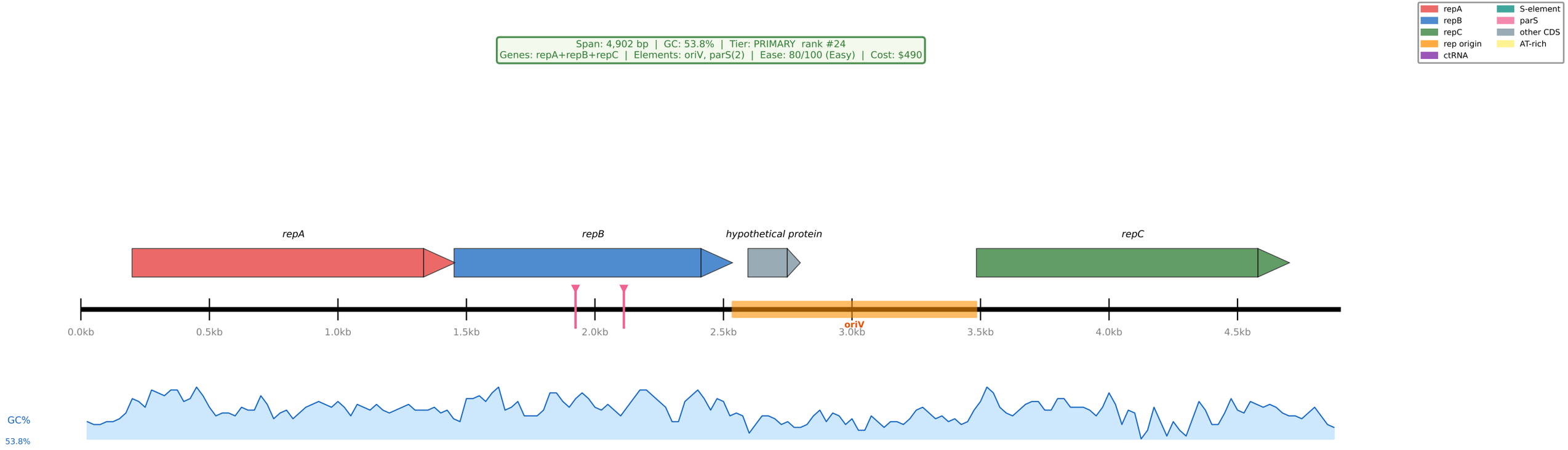
1D1514_plasmid2_pTi_Type_V_rep_2 | Type V rep 2 | Agrobacterium tumefaciens

Span: 4,152 bp | GC: 52.3% | Tier: PRIMARY rank #15
Genes: repA+repB+repC | Elements: parS(2) | Ease: 80/100 (Easy) | Cost: $415



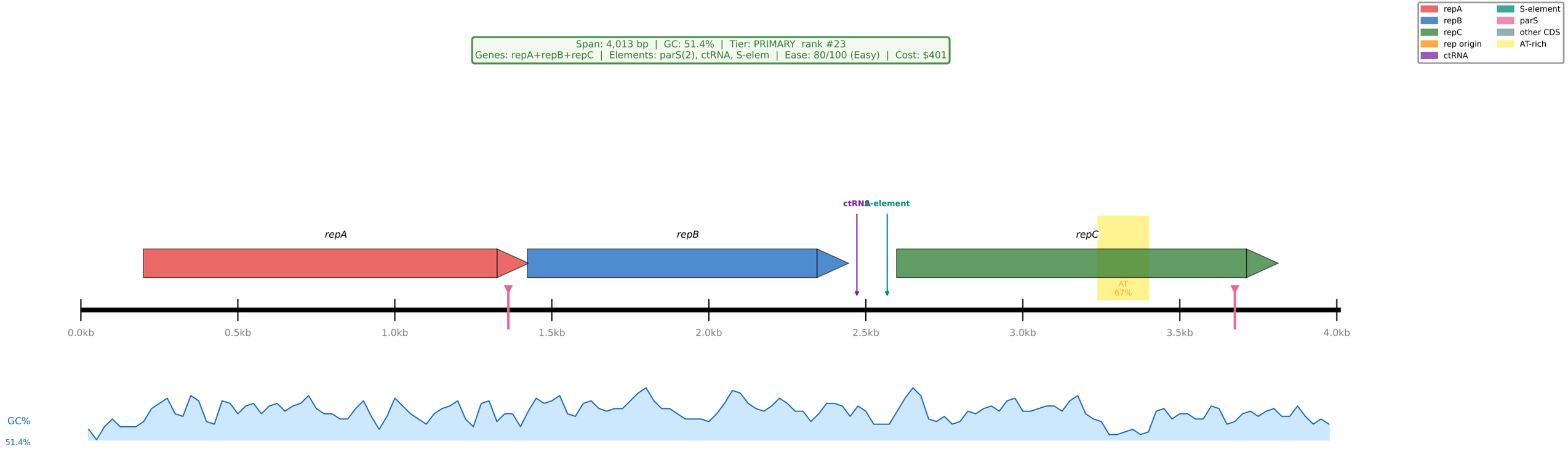
1D1524_chromid | chromid | Agrobacterium tumefaciens

Span: 4,902 bp | GC: 53.8% | Tier: PRIMARY rank #24
Genes: repA+repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $490



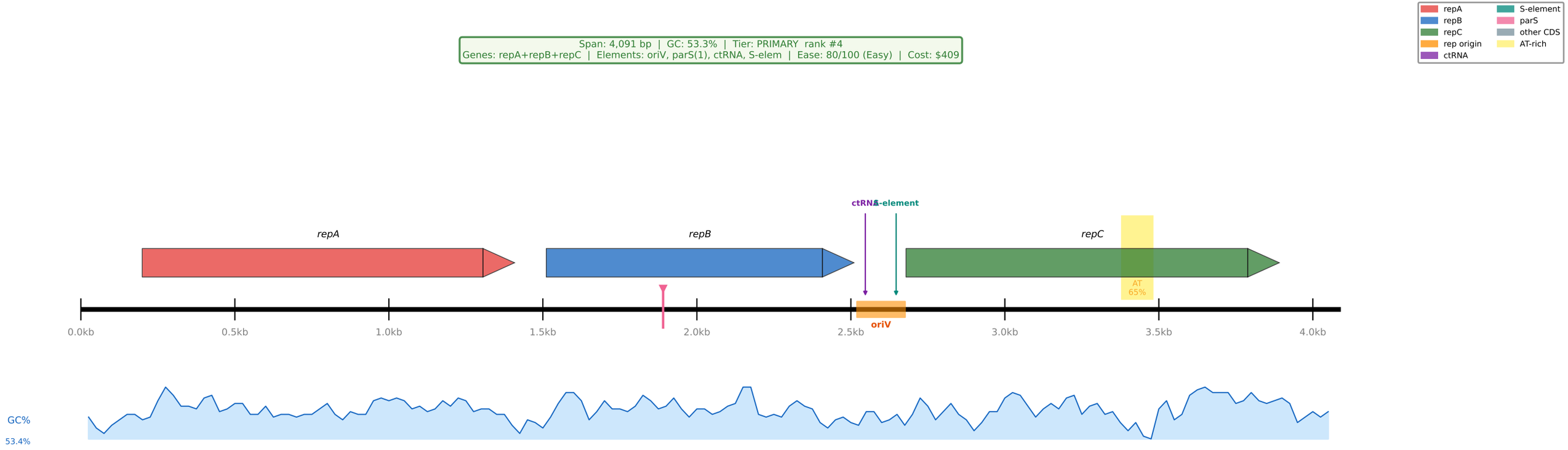
1D1524_pAt | pAt | Agrobacterium tumefaciens

Span: 4,013 bp | GC: 51.4% | Tier: PRIMARY rank #23
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



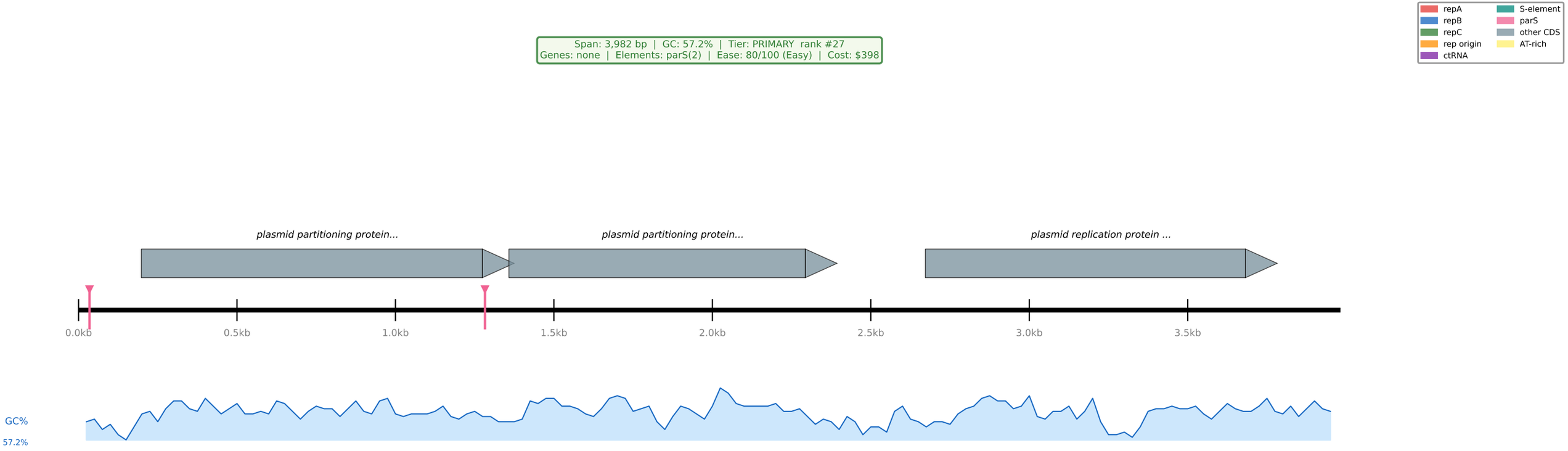
AF3.44_contig_3_pTi_Type_X | Type X | Agrobacterium tumefaciens

Span: 4,091 bp | GC: 53.3% | Tier: PRIMARY rank #4
Genes: repA+repB+repC | Elements: oriV, parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $409



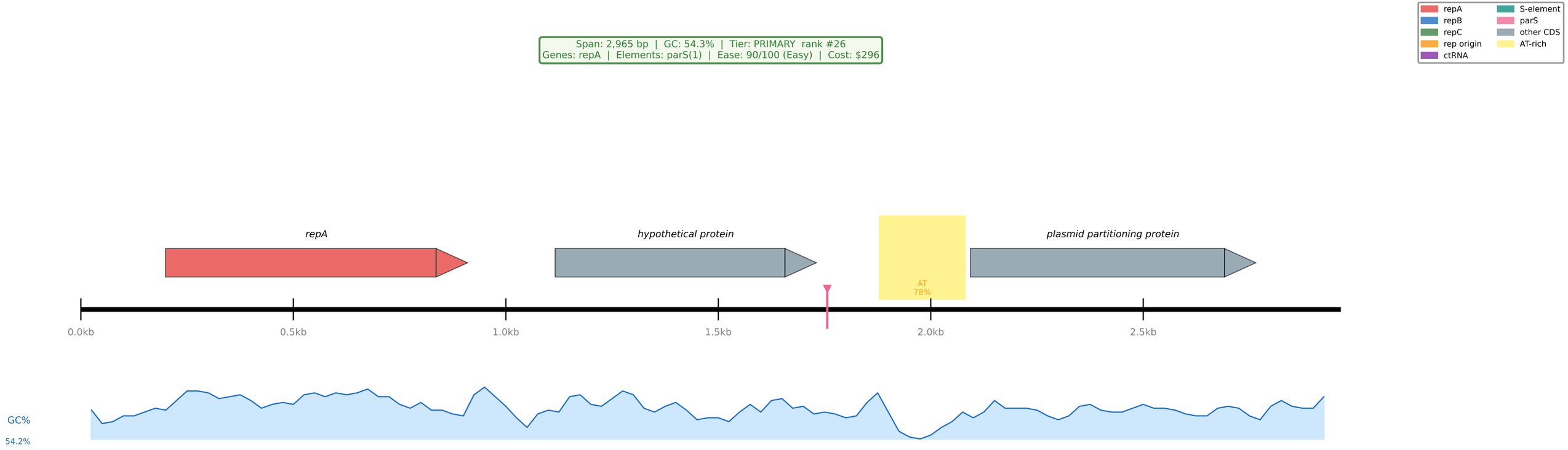
Atu_K84_chromid_CP000629.1 | chromid | Agrobacterium radiobacter

Span: 3,982 bp | GC: 57.2% | Tier: PRIMARY rank #27
Genes: none | Elements: parS(2) | Ease: 80/100 (Easy) | Cost: $398



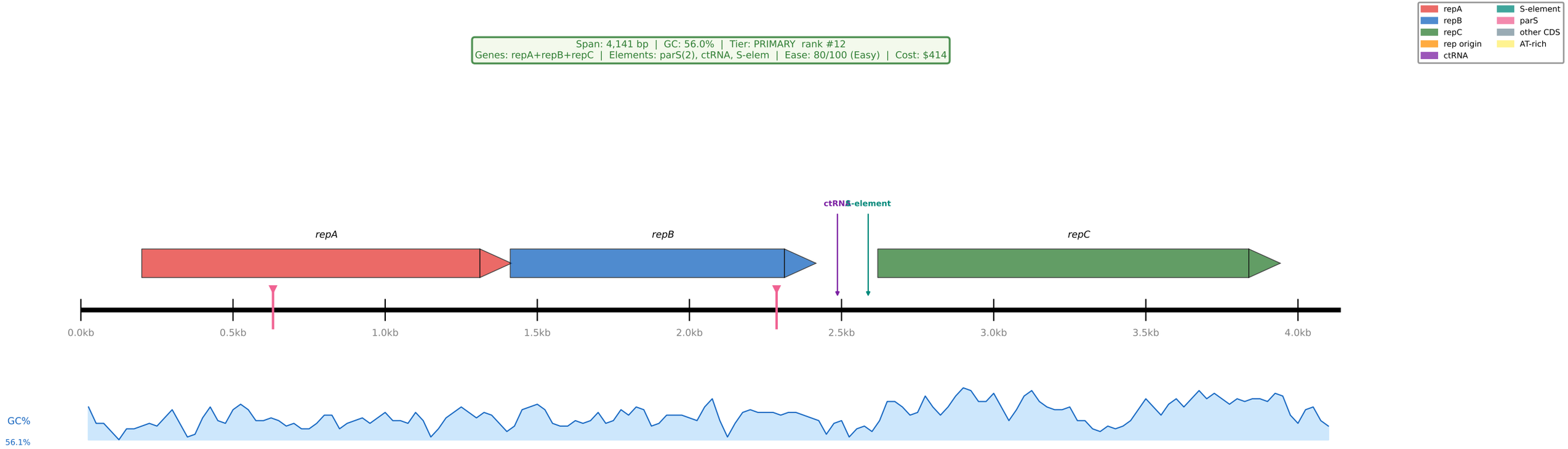
Atu_K84_pAgK84_CP000632.1 | pAg | Agrobacterium radiobacter

Span: 2,965 bp | GC: 54.3% | Tier: PRIMARY rank #26
Genes: repA | Elements: parS(1) | Ease: 90/100 (Easy) | Cost: $296



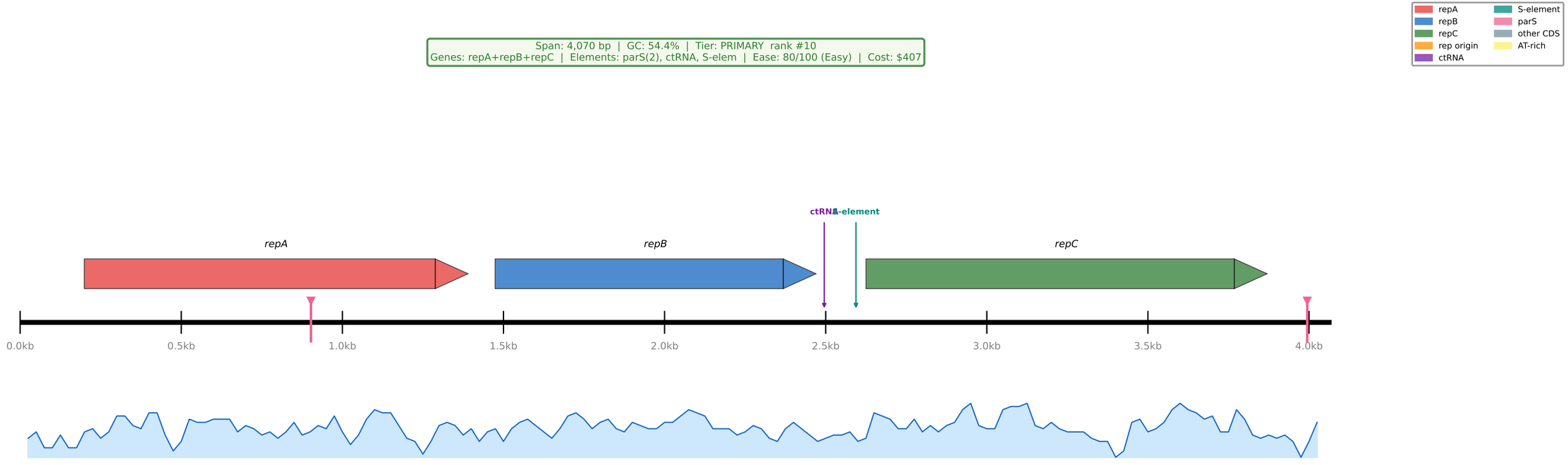
B21_90_contig_3_pTi_Type_VII | Type VII | Agrobacterium tumefaciens

Span: 4,141 bp | GC: 56.0% | Tier: PRIMARY rank #12
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $414



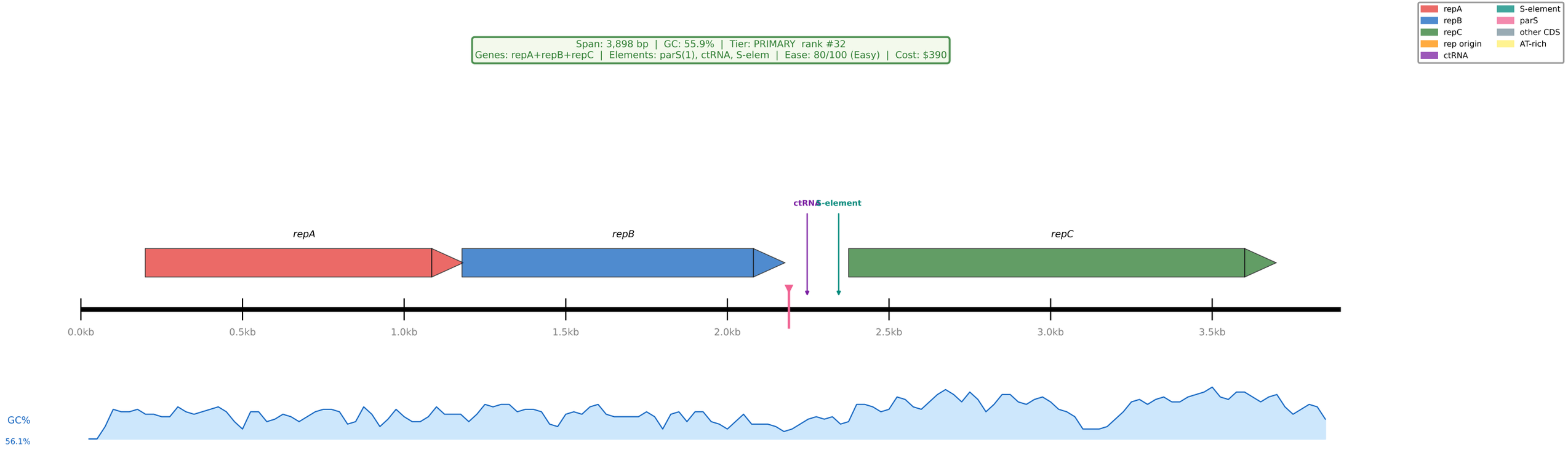
B230_85_contig_5_pTi_Type_VIII.a | Type VIII.a | Agrobacterium tumefaciens

Span: 4,070 bp | GC: 54.4% | Tier: PRIMARY rank #10
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $407



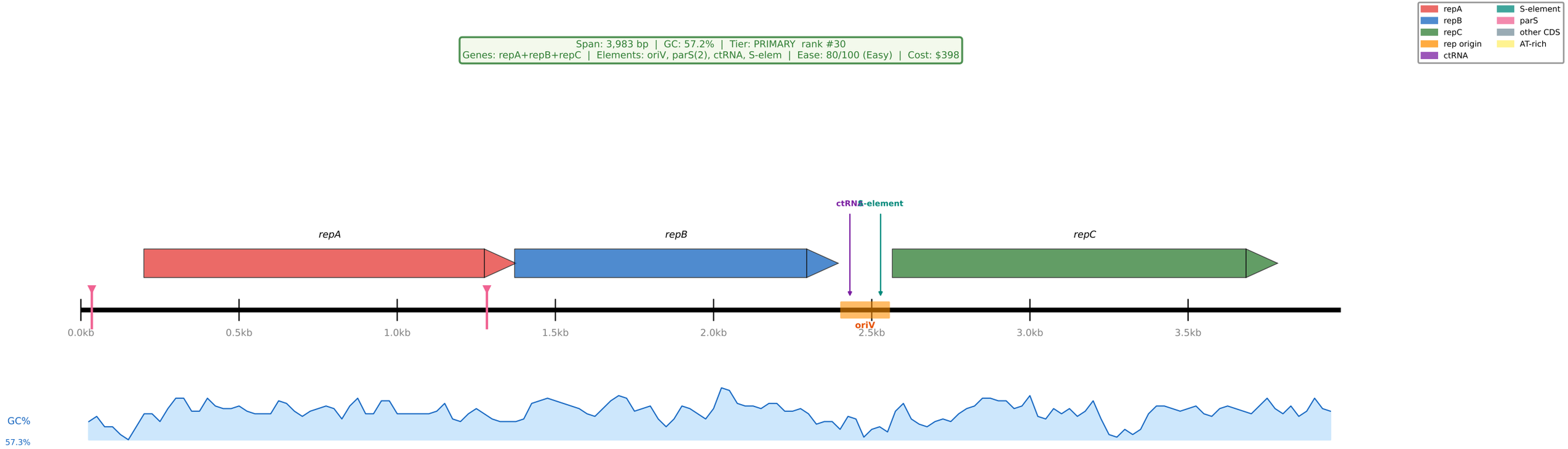
B230_85_contig_6 | unknown | Agrobacterium tumefaciens

Span: 3,898 bp | GC: 55.9% | Tier: PRIMARY rank #32
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $390



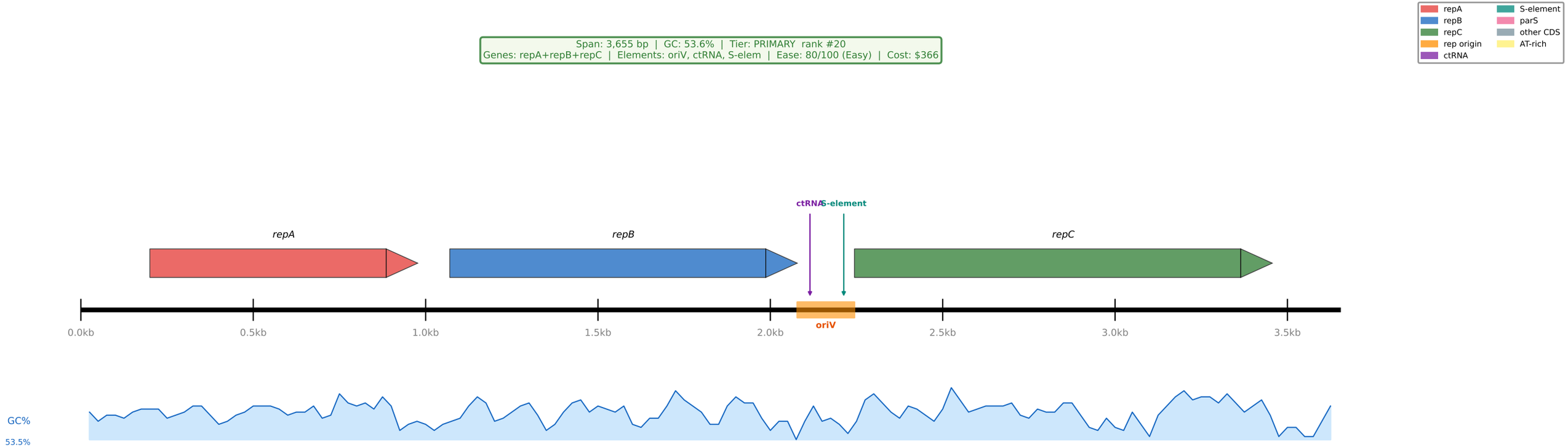
CA75_95_contig_2 | unknown | Agrobacterium tumefaciens

Span: 3,983 bp | GC: 57.2% | Tier: PRIMARY rank #30
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



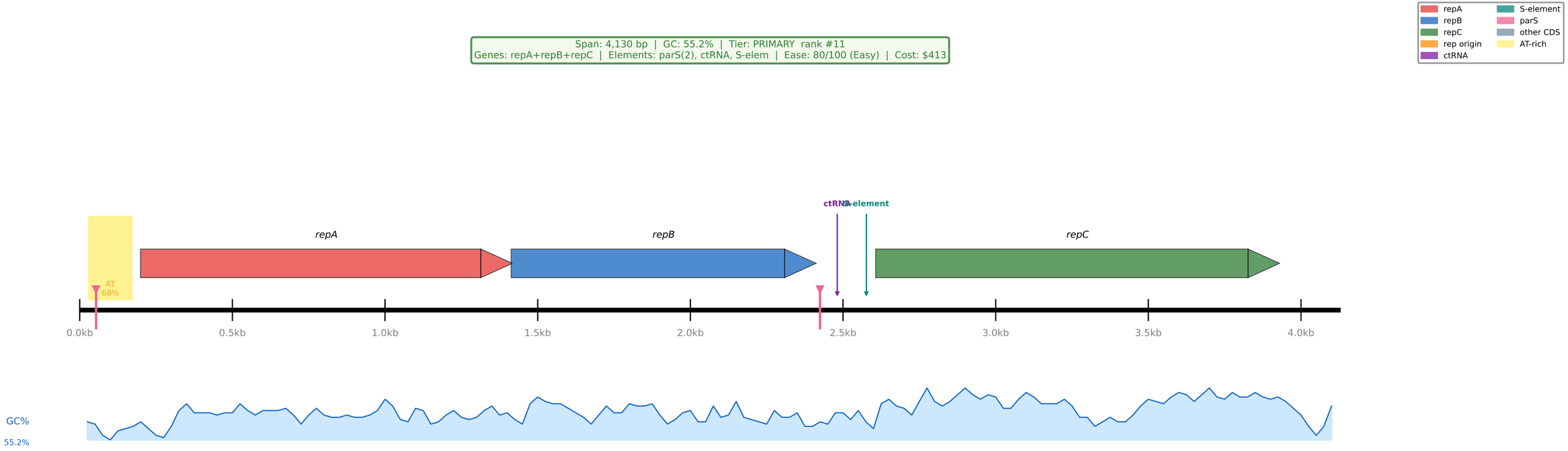
K599_CAICSX020000003.1_pRi_Type_I | Type I | Agrobacterium rhizogenes

Span: 3,655 bp | GC: 53.6% | Tier: PRIMARY rank #20
Genes: repA+repB+repC | Elements: oriV, ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $366



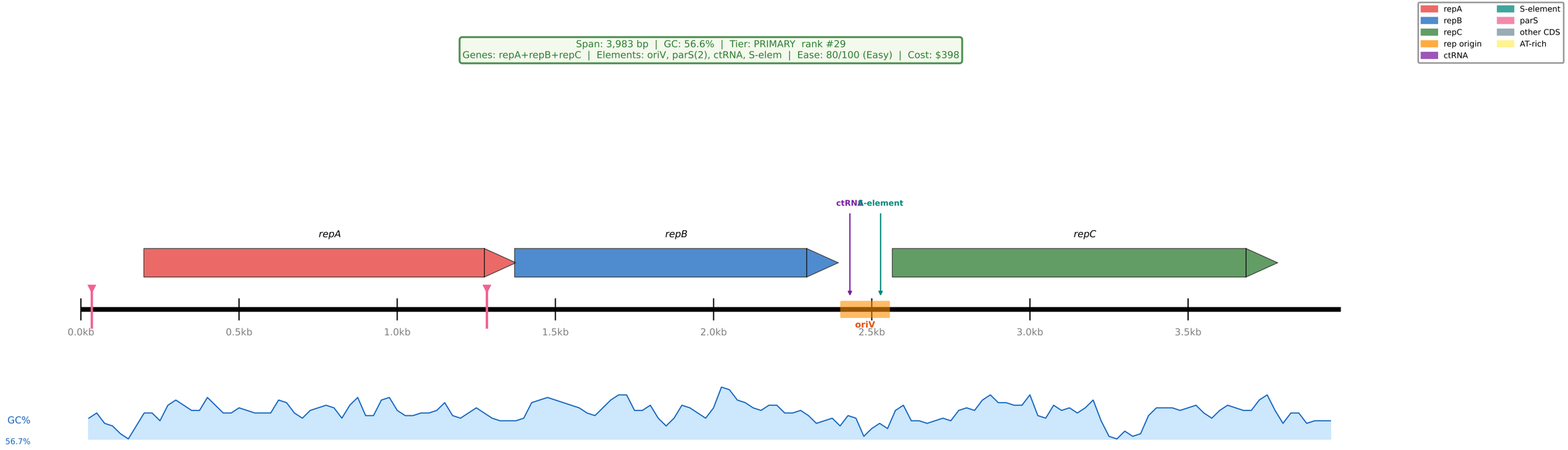
L51_94_contig_5_pTi_Type_VIII.b | Type VIII.b | Agrobacterium tumefaciens

Span: 4,130 bp | GC: 55.2% | Tier: PRIMARY rank #11
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $413



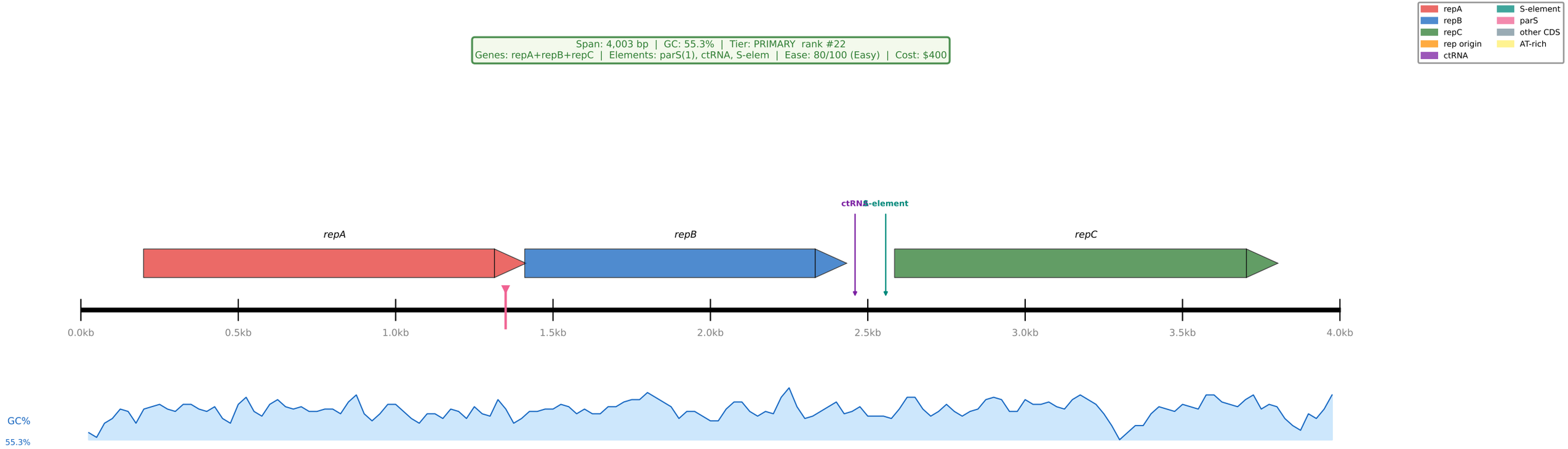
NCIB_8196_NZ_JAAMDI010000039.1 | unknown | Agrobacterium rhizogenes

Span: 3,983 bp | GC: 56.6% | Tier: PRIMARY rank #29
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



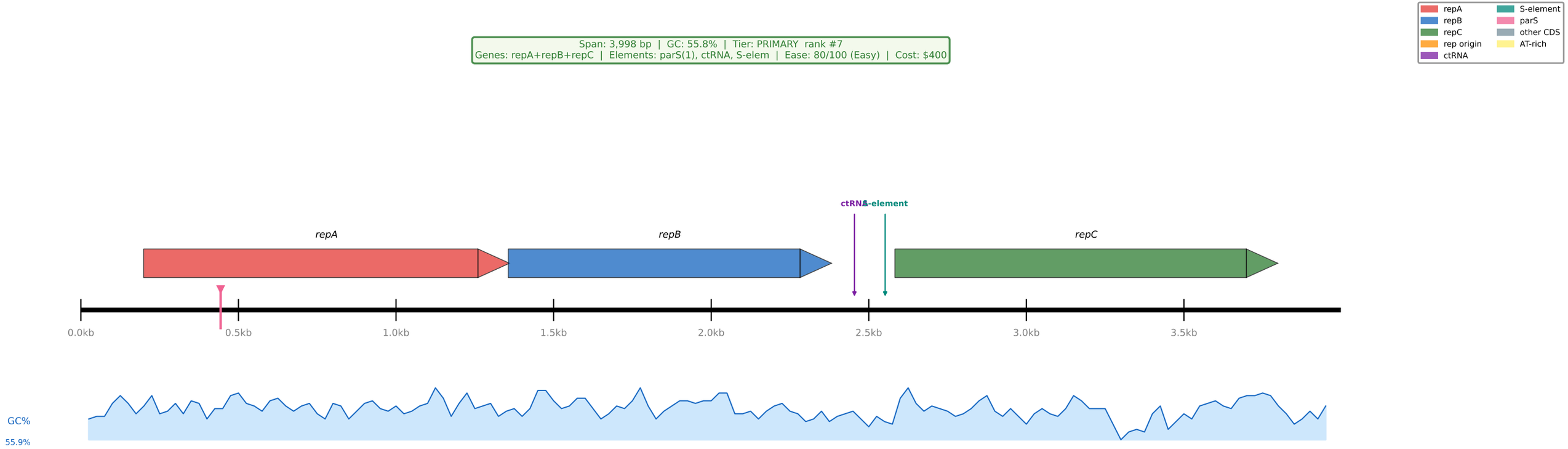
NCIB_8196_NZ_JAAMDI010000041.1_pRi_Type_II | Type II | Agrobacterium rhizogenes

Span: 4,003 bp | GC: 55.3% | Tier: PRIMARY rank #22
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



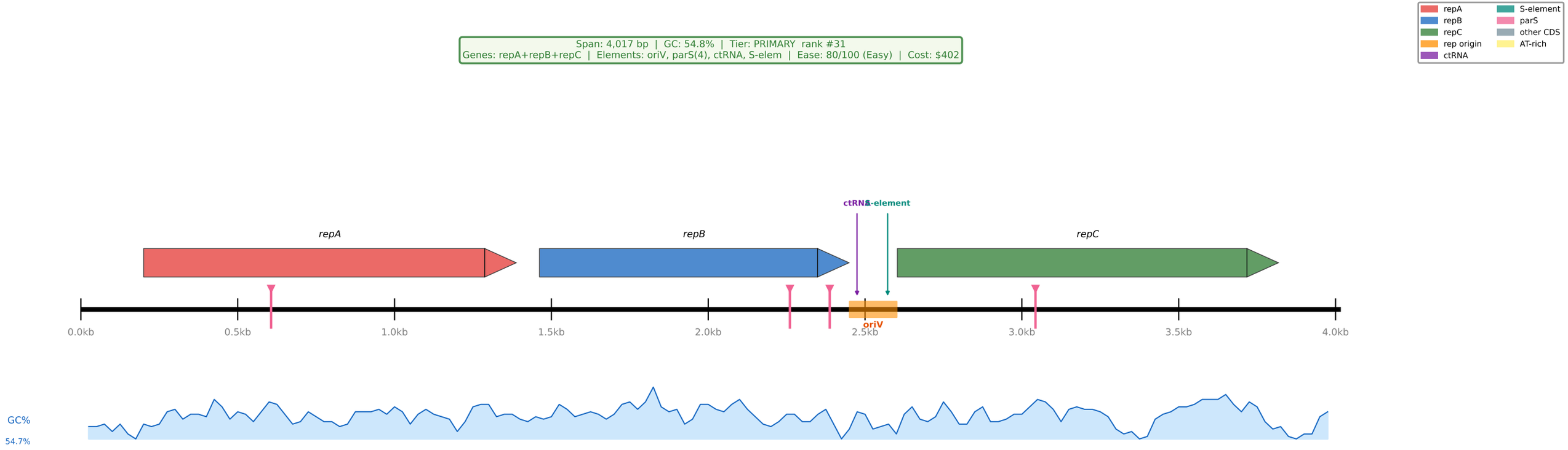
NCPPB_1641_NZ_LT009777.1_pTi_Type_XI | Type XI | Agrobacterium vitis

Span: 3,998 bp | GC: 55.8% | Tier: PRIMARY rank #7
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



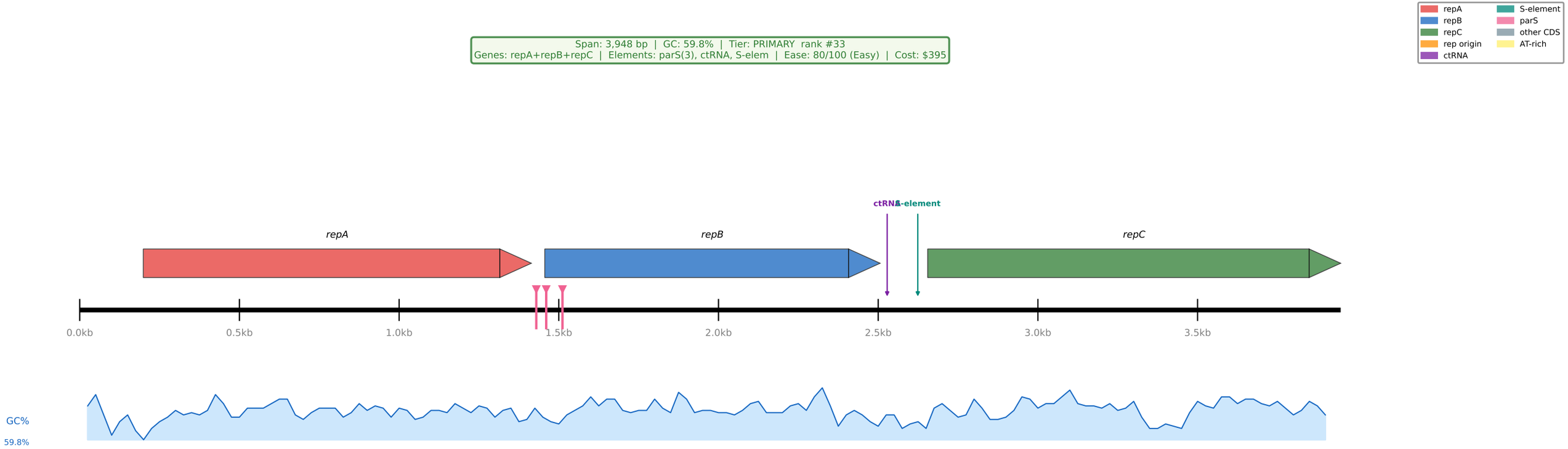
Q15_94_contig_6 | unknown | Agrobacterium tumefaciens

Span: 4,017 bp | GC: 54.8% | Tier: PRIMARY rank #31
Genes: repA+repB+repC | Elements: oriV, parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



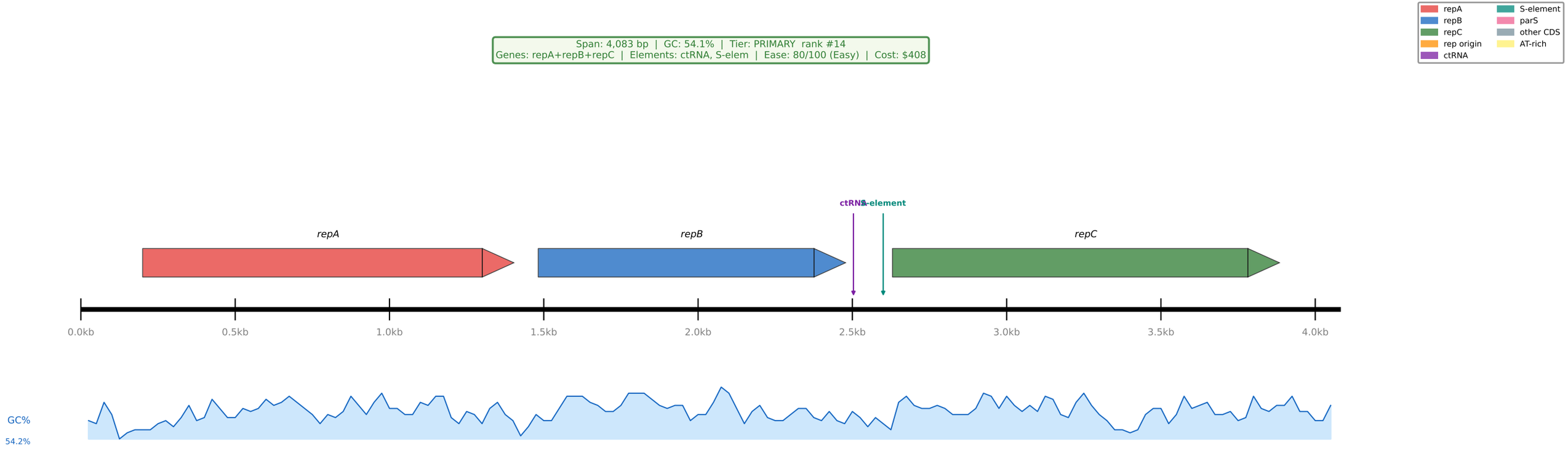
Rhizobium_etli_CFN_42_CP000135.1 | unknown | Rhizobium etli CFN 42

Span: 3,948 bp | GC: 59.8% | Tier: PRIMARY rank #33
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $395



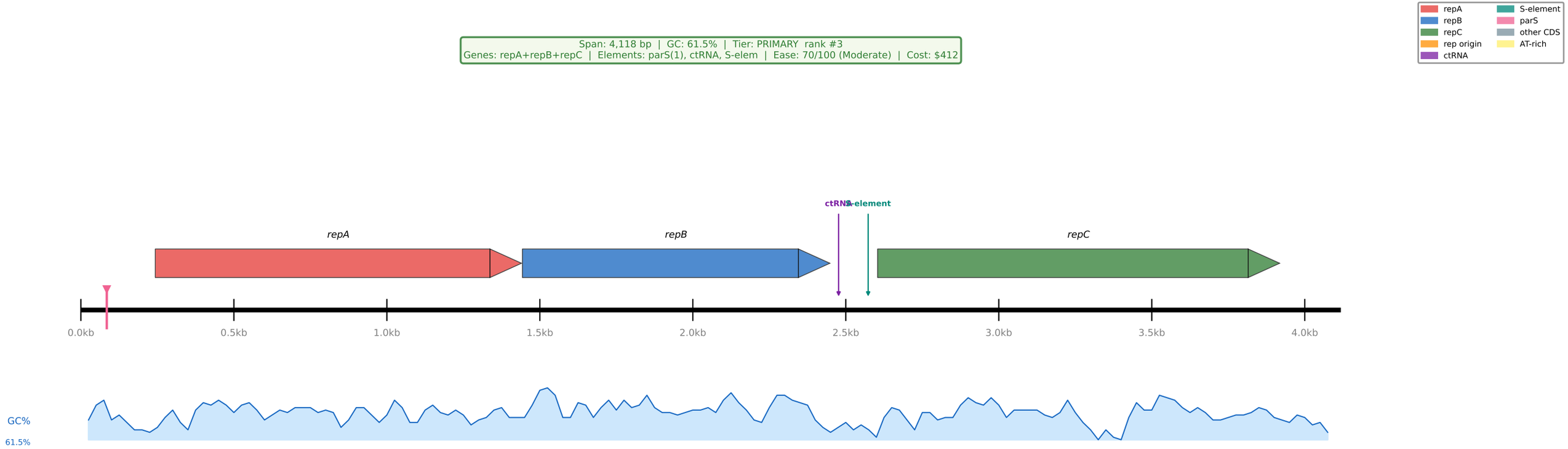
S7_73_contig_3_pTi_Type_IX | Type IX | Agrobacterium tumefaciens

Span: 4,083 bp | GC: 54.1% | Tier: PRIMARY rank #14
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $408



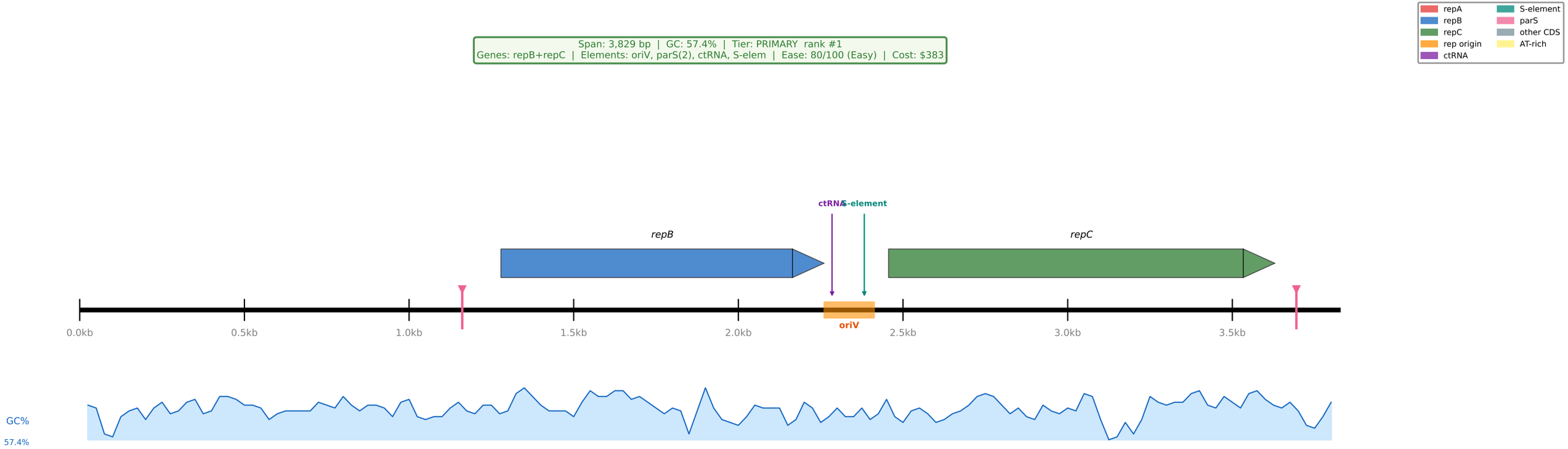
Sinorhizobium_fredii_NGR234_CP000874.1 | unknown | Sinorhizobium fredii NGR234

Span: 4,118 bp | GC: 61.5% | Tier: PRIMARY rank #3
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $412



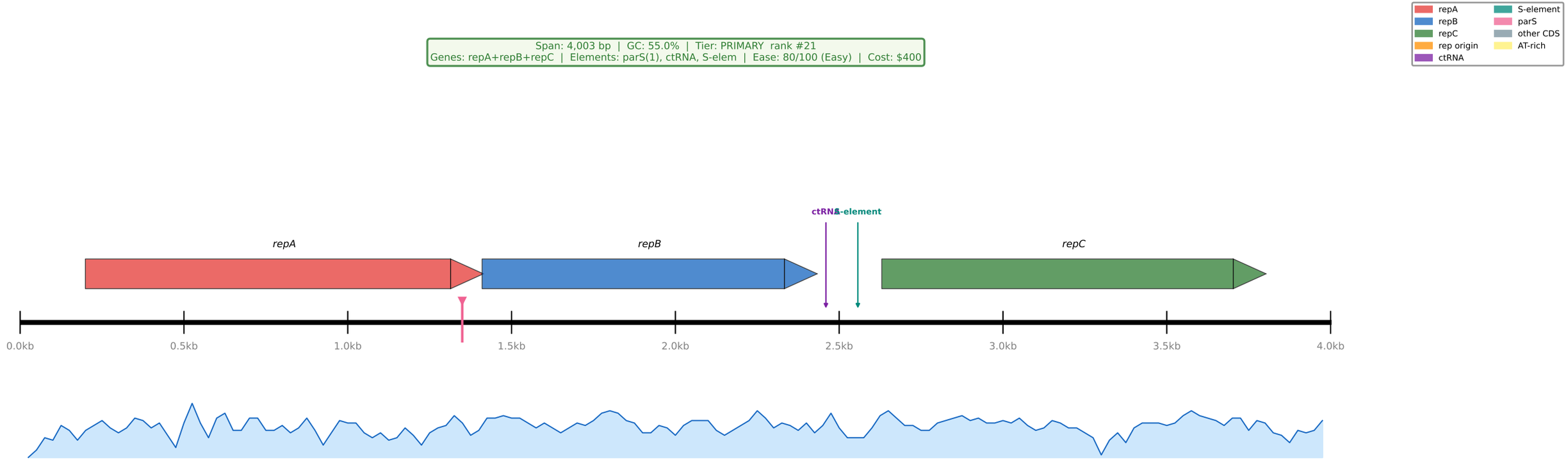
Sinorhizobium_fredii_NGR234_U00090.2 | unknown | Sinorhizobium fredii NGR234

Span: 3,829 bp | GC: 57.4% | Tier: PRIMARY rank #1
Genes: repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $383



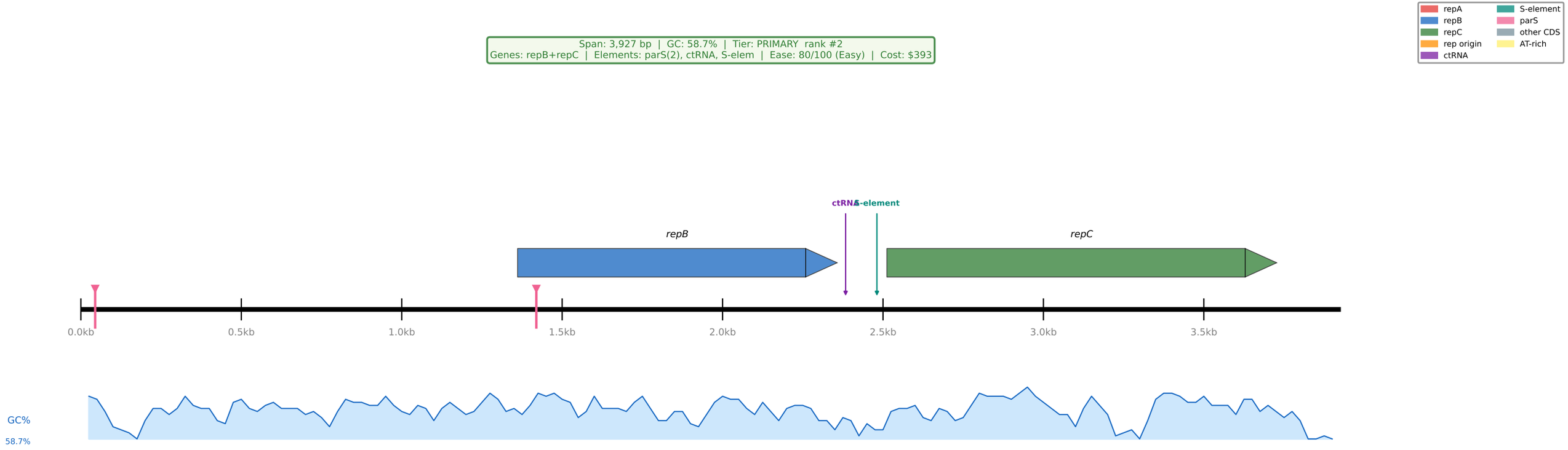
T155_95_NZ_JAAMCX010000012.1_pRi_Type_III | Type III | Agrobacterium rhizogenes

Span: 4,003 bp | GC: 55.0% | Tier: PRIMARY rank #21
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



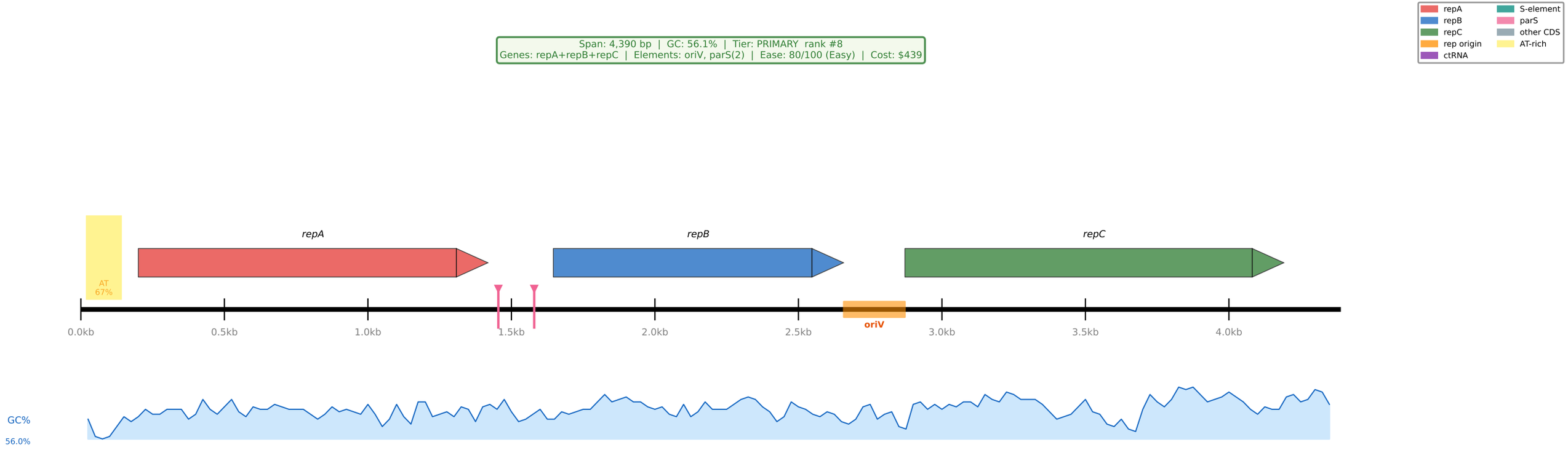
WS163_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,927 bp | GC: 58.7% | Tier: PRIMARY rank #2
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $393



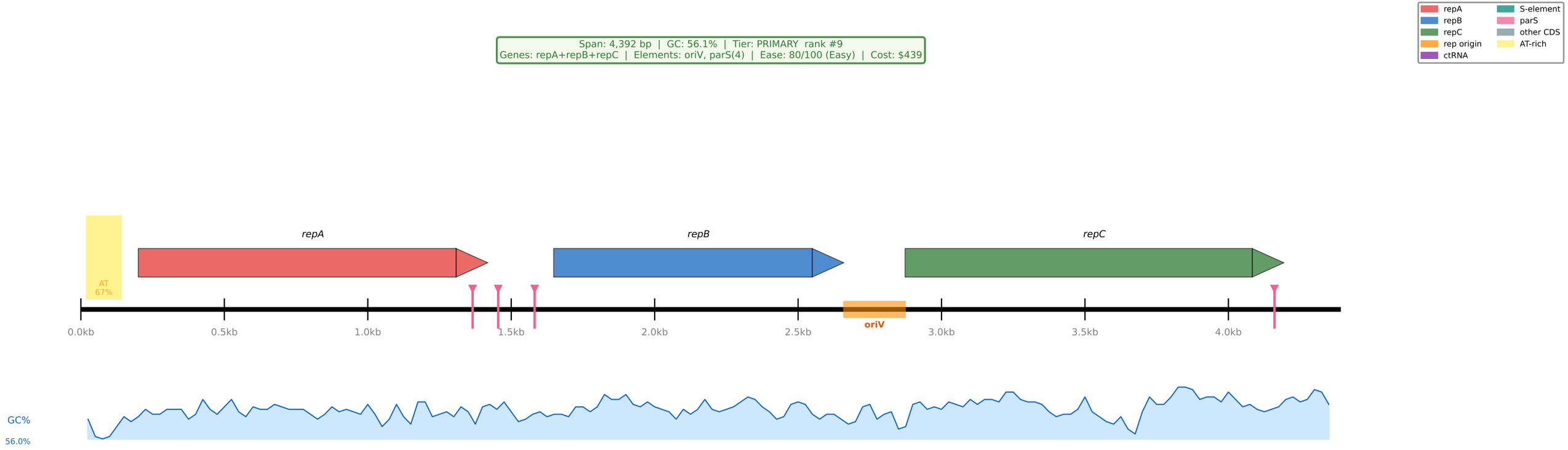
1D1524_plasmid2_pTi_Type_IV.a | Type IV.a | Agrobacterium tumefaciens

Span: 4,390 bp | GC: 56.1% | Tier: PRIMARY rank #8
Genes: repA+repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $439



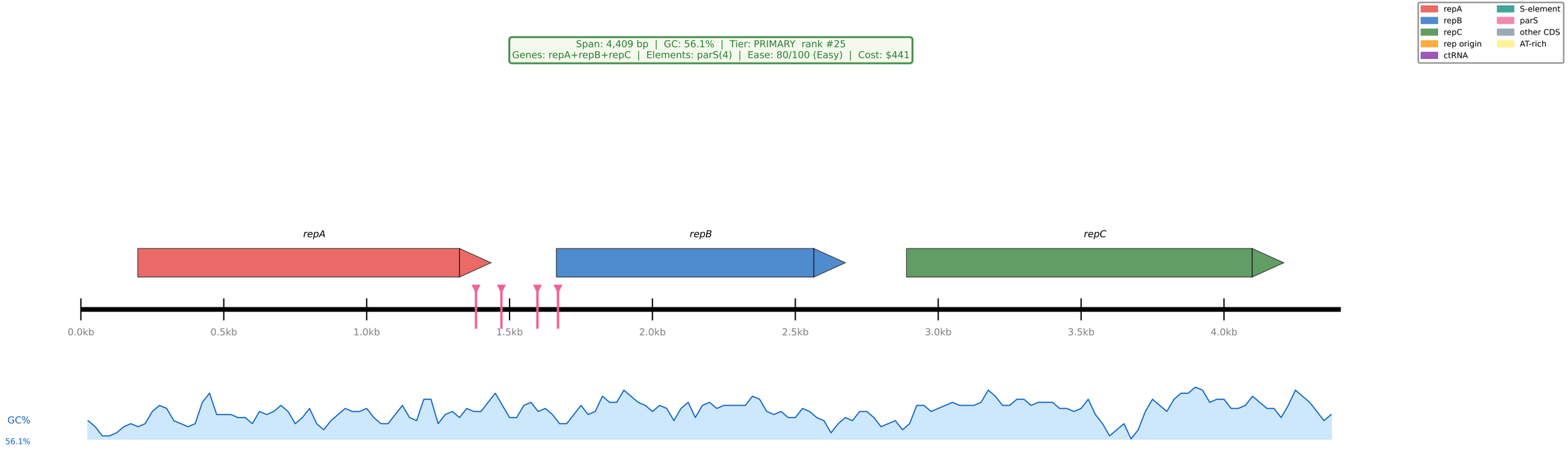
AMS008_plasmid3_pTi_Type_IV.b | Type IV.b | Agrobacterium tumefaciens

Span: 4,392 bp | GC: 56.1% | Tier: PRIMARY rank #9
Genes: repA+repB+repC | Elements: oriV, parS(4) | Ease: 80/100 (Easy) | Cost: $439



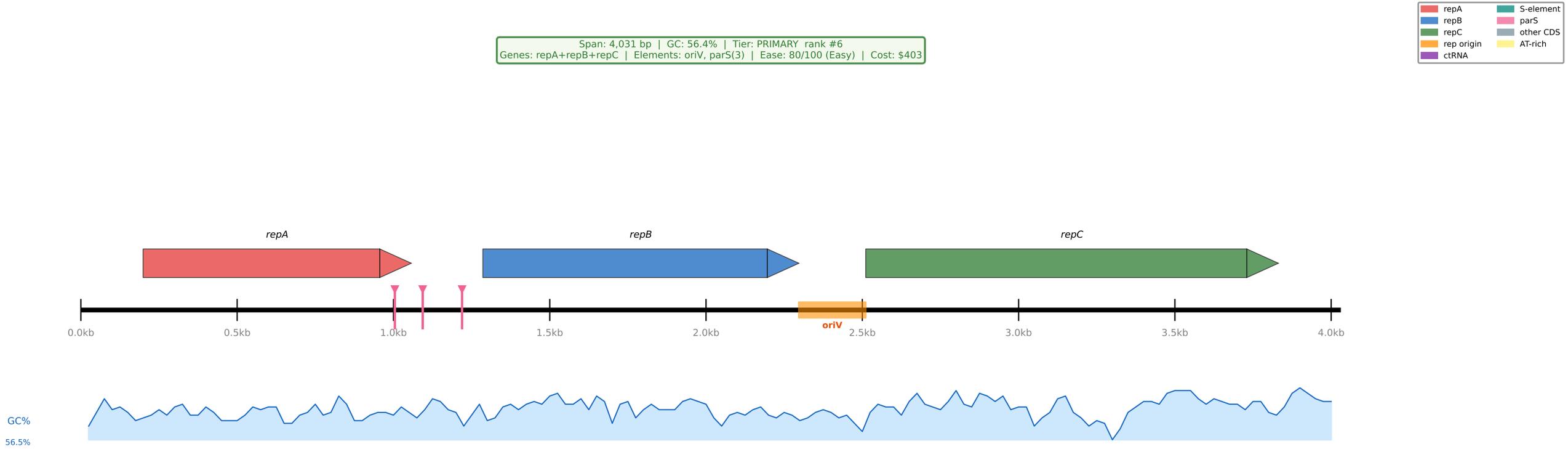
Atu_K84_pAtK84b_CP000630.1 | pAt | Agrobacterium radiobacter

Span: 4,409 bp | GC: 56.1% | Tier: PRIMARY rank #25
Genes: repA+repB+repC | Elements: parS(4) | Ease: 80/100 (Easy) | Cost: $441



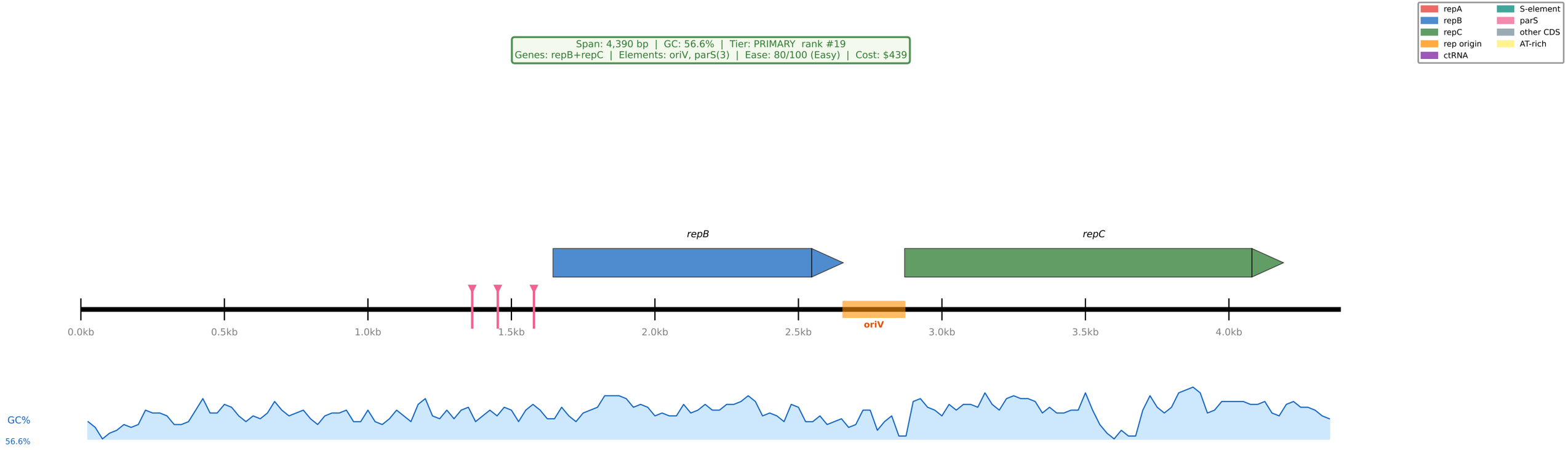
C58_AE007871.2_pTi_Type_I.a | Type I.a | Agrobacterium fabrum

Span: 4,031 bp | GC: 56.4% | Tier: PRIMARY rank #6
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $403



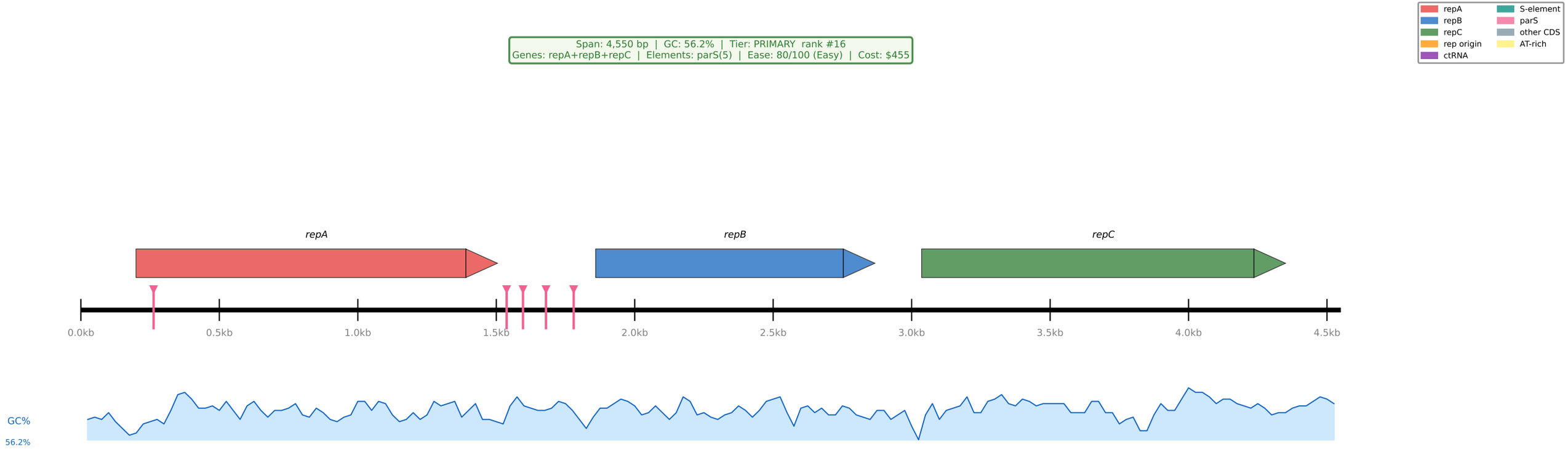
CA75_95_contig_3_pTi_Type_IV.c | Type IV.c | Agrobacterium tumefaciens

Span: 4,390 bp | GC: 56.6% | Tier: PRIMARY rank #19
Genes: repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $439



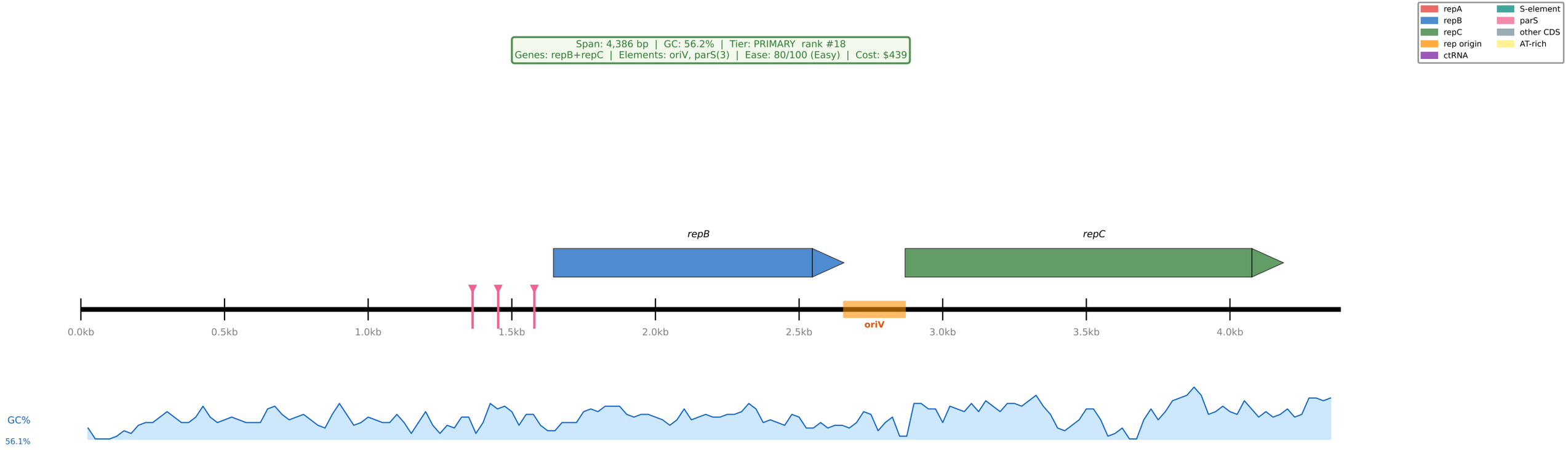
CG474_CG474_3_pTi_Type_I.c | Type I.c | Agrobacterium tumefaciens

Span: 4,550 bp | GC: 56.2% | Tier: PRIMARY rank #16
Genes: repA+repB+repC | Elements: parS(5) | Ease: 80/100 (Easy) | Cost: $455



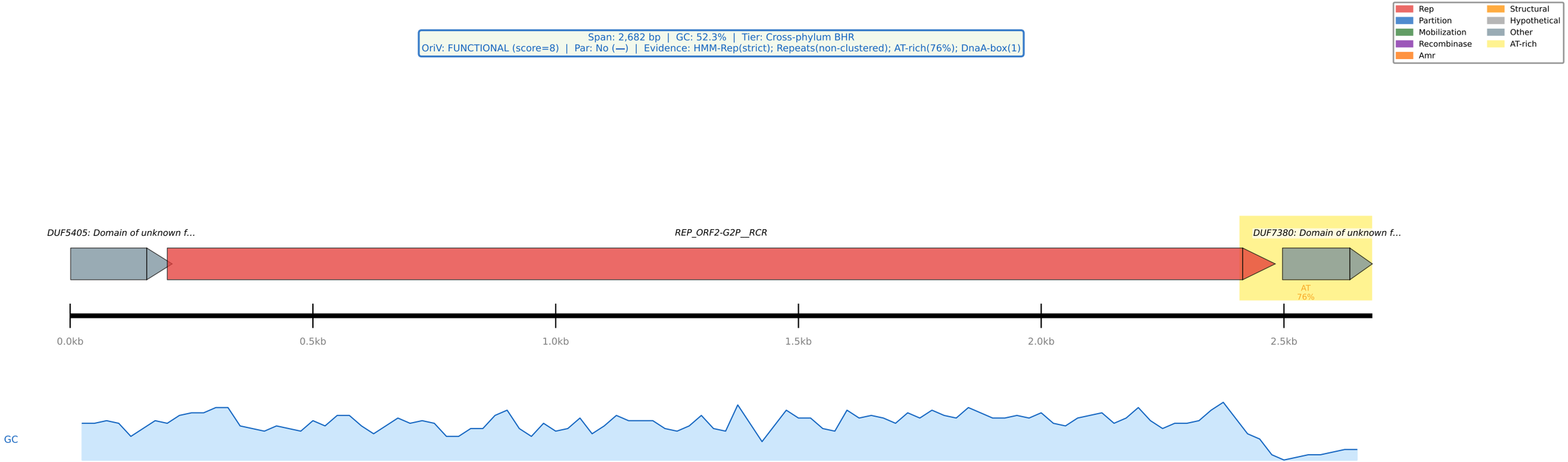
Q15_94_contig_5_pTi_Type_VI | Type VI | Agrobacterium tumefaciens

Span: 4,386 bp | GC: 56.2% | Tier: PRIMARY rank #18
Genes: repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $439



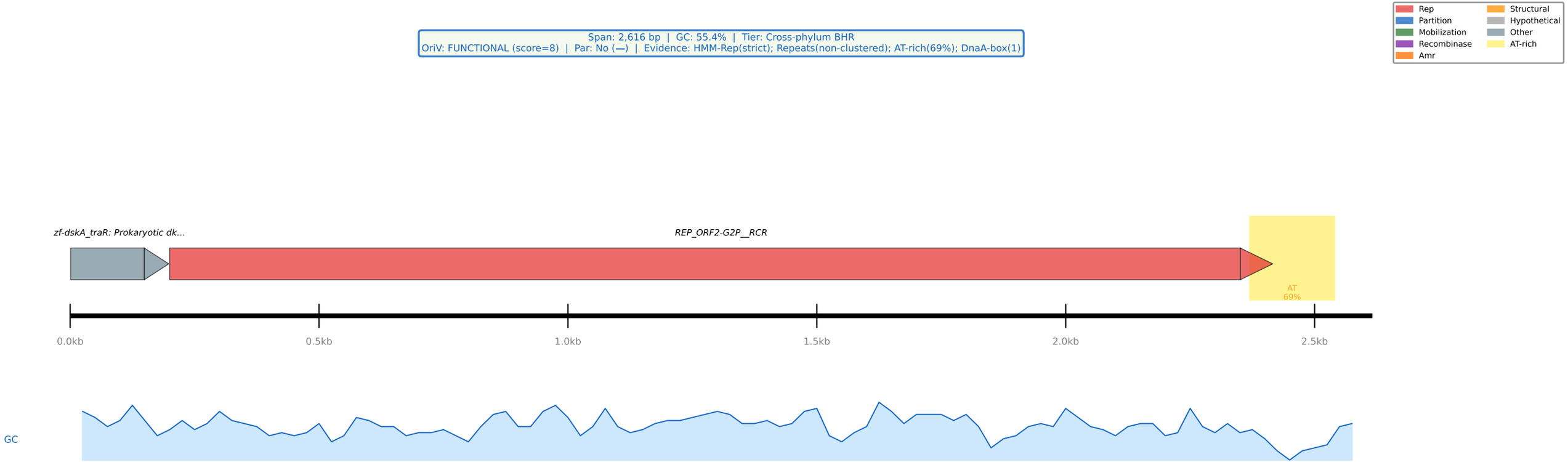
NZ_CP019203.1 | REP_ORF2-G2P__RCR | Salmonella_enterica

Span: 2,682 bp | GC: 52.3% | Tier: Cross-phylum BHR
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered); AT-rich(76%); DnaA-box(1)



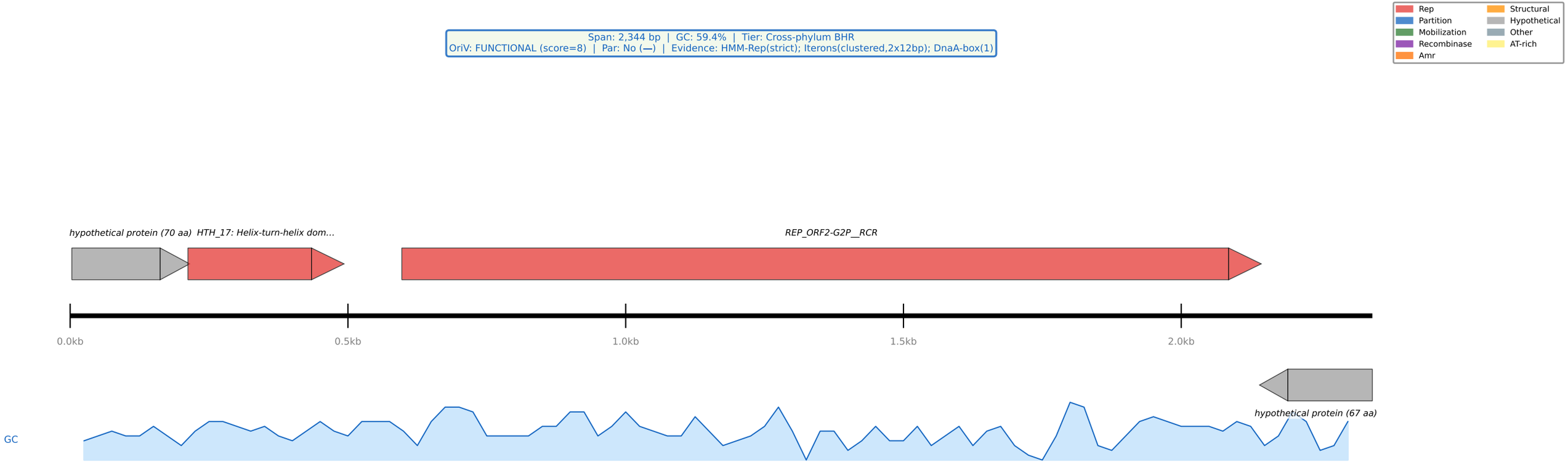
NZ_CP058189.1 | REP_ORF2-G2P__RCR | Enterobacter_hormaechei

Span: 2,616 bp | GC: 55.4% | Tier: Cross-phylum BHR
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered); AT-rich(69%); DnaA-box(1)



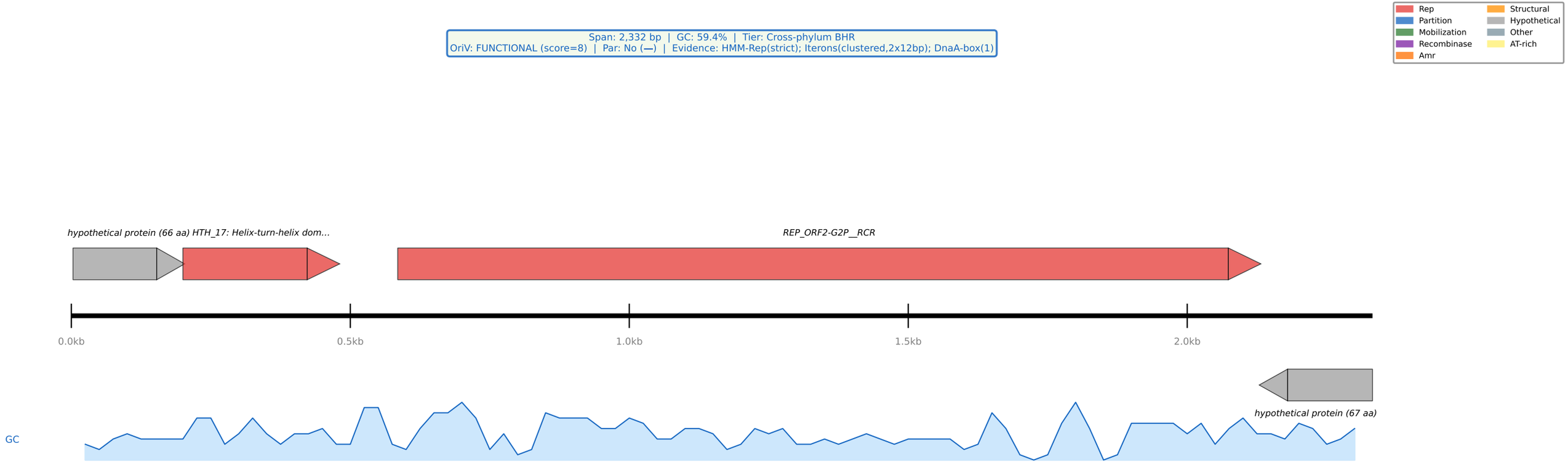
NZ_LC331663.1 | REP_ORF2-G2P__RCR | Rhodococcus_erythropolis

Span: 2,344 bp | GC: 59.4% | Tier: Cross-phylum BHR
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp); DnaA-box(1)



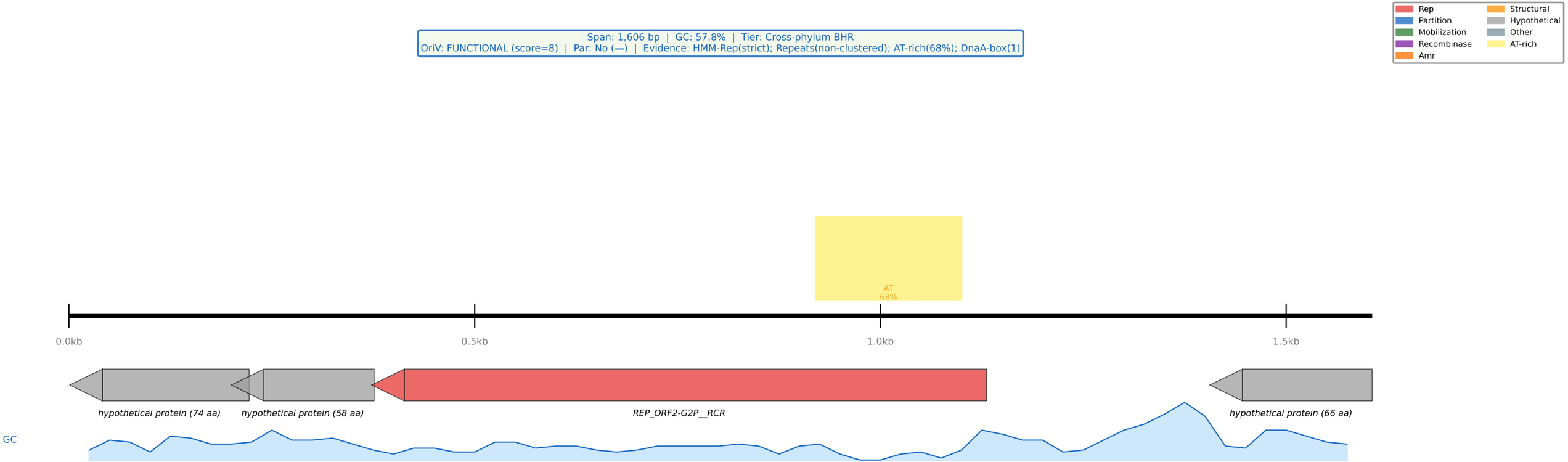
NZ_AP018736.1 | REP_ORF2-G2P__RCR | Rhodococcus_erythropolis

Span: 2,332 bp | GC: 59.4% | Tier: Cross-phylum BHR
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp); DnaA-box(1)



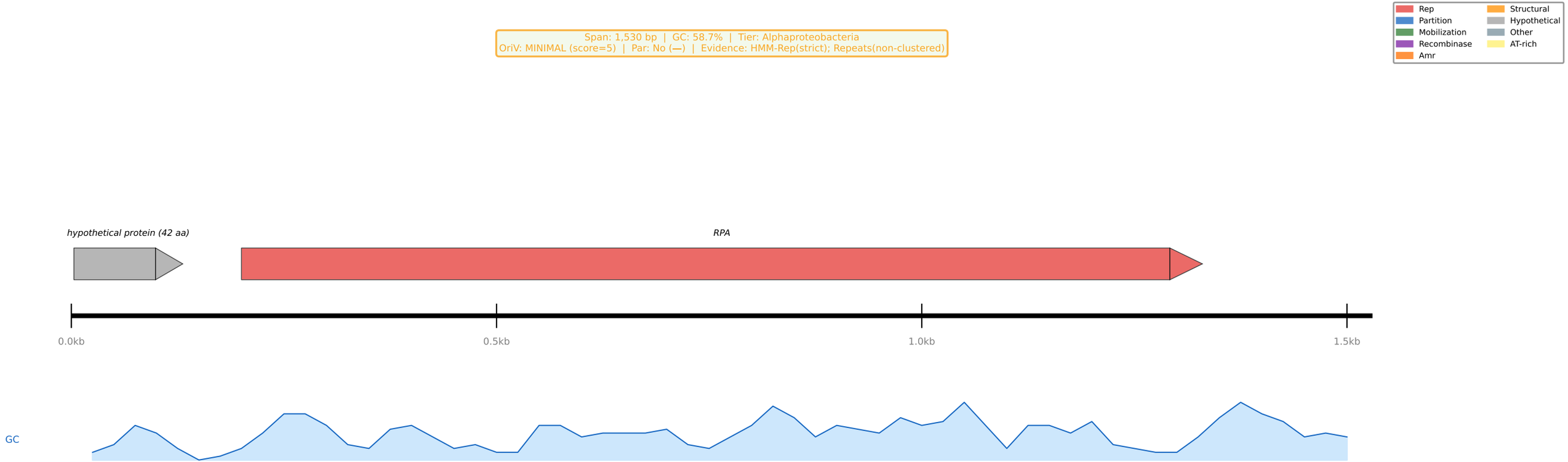
OY725836.1 | REP_ORF2-G2P__RCR | Subdoligranulum_variabile

Span: 1,606 bp | GC: 57.8% | Tier: Cross-phylum BHR
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered); AT-rich(68%); DnaA-box(1)



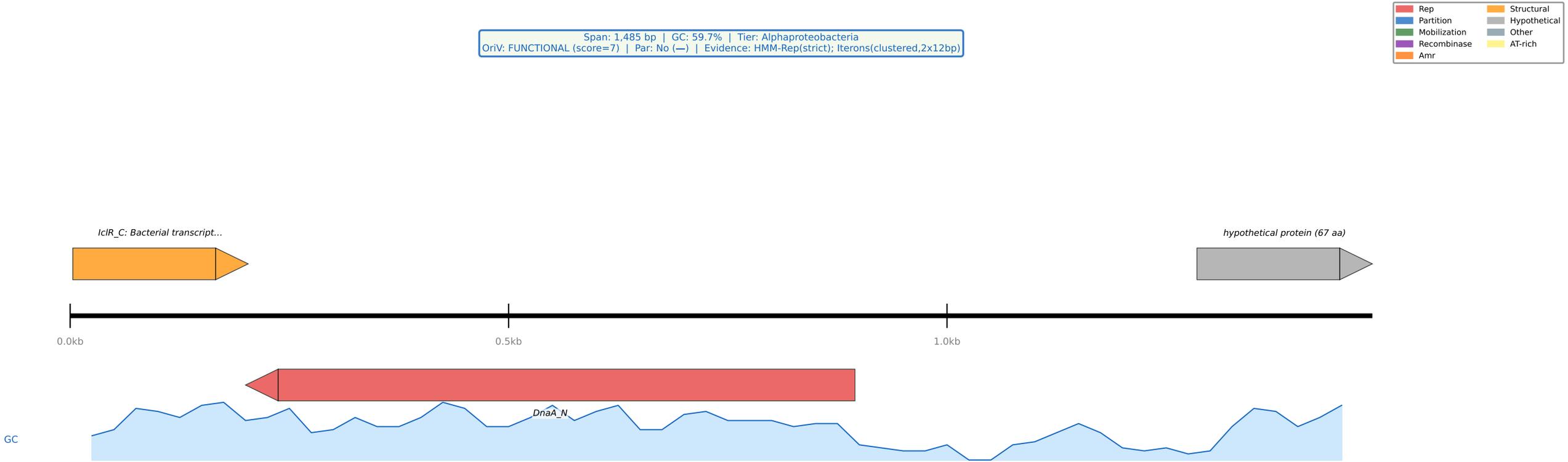
NZ_CP015988.1 | RPA | Sphingobium_sp._EP60837

Span: 1,530 bp | GC: 58.7% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



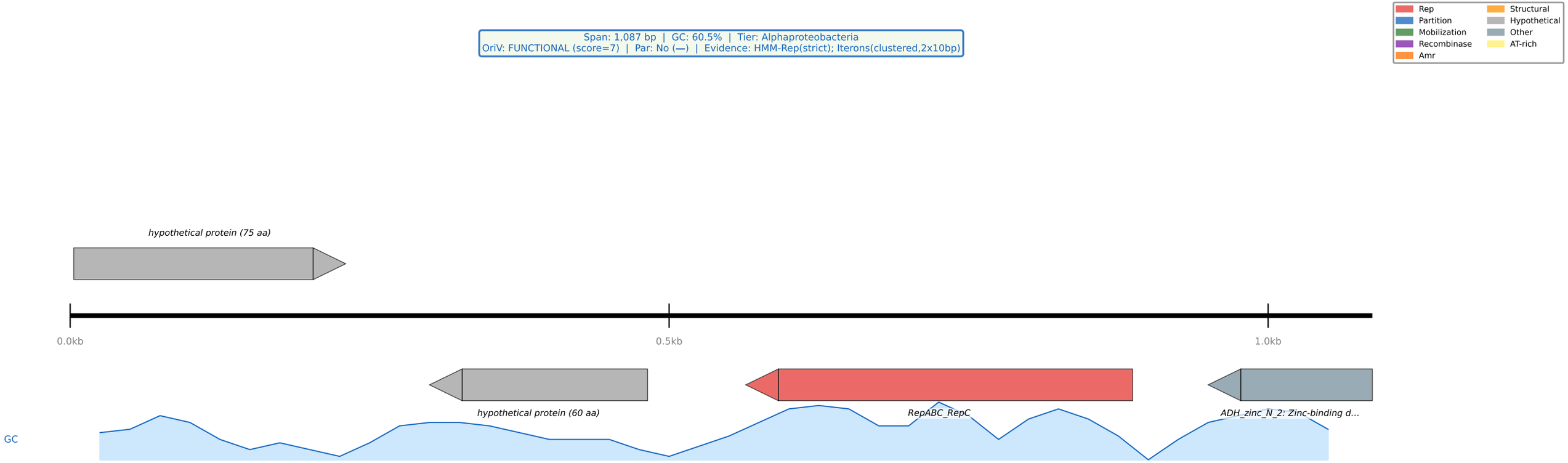
NZ_CP025809.1 | DnaA_N | Sulfitobacter_sp._SK025

Span: 1,485 bp | GC: 59.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)

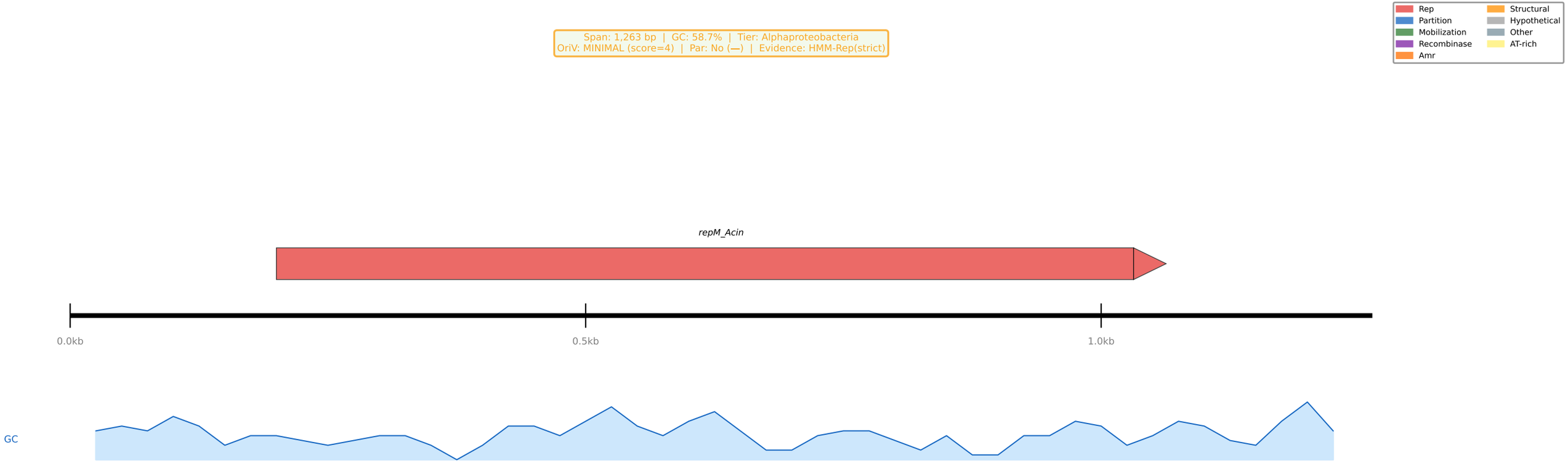


NZ_CP036517.1 | PRK13824.1 | Lichenihabitans_psoromatis

Span: 1,087 bp | GC: 60.5% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp)

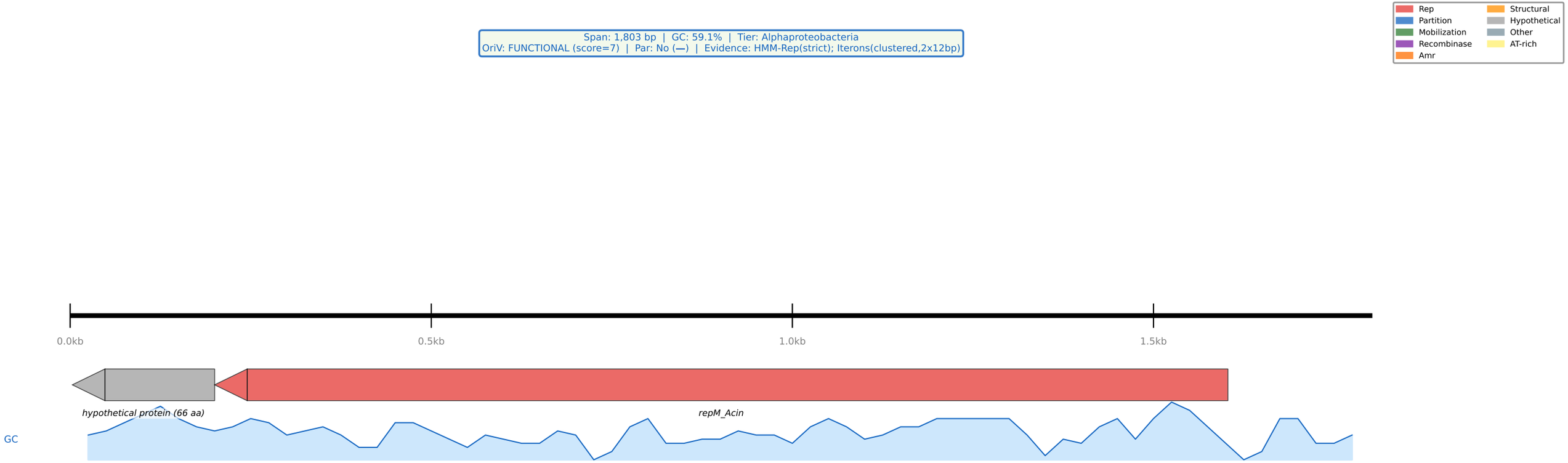


NZ_CP044072.1 | repM_Acin | Pseudomonas_oryzihabitans



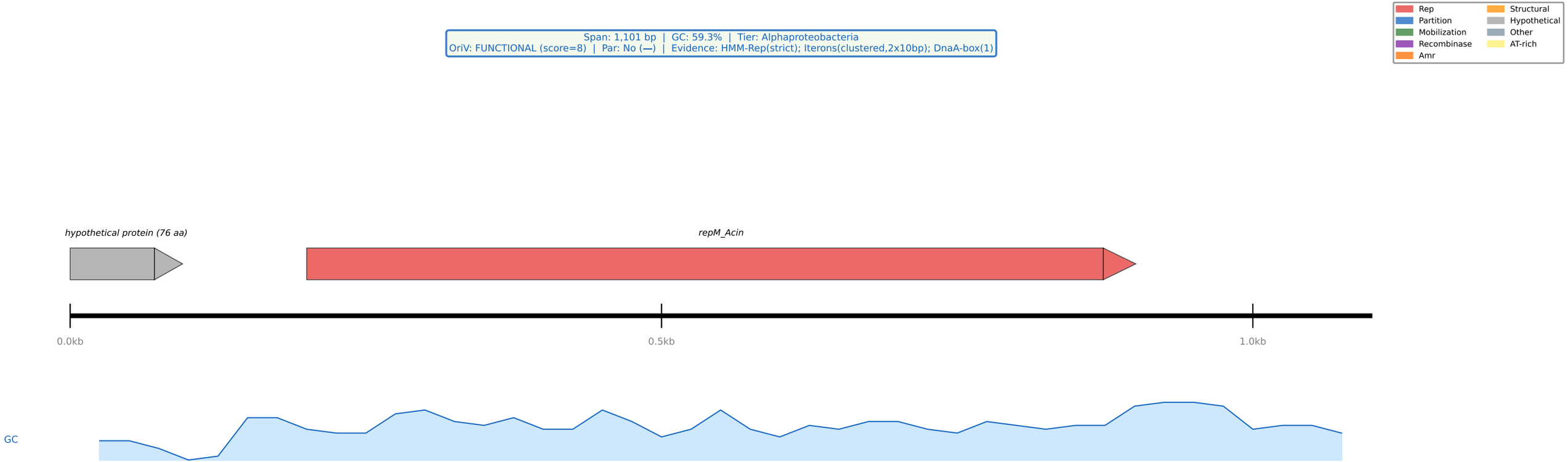
NZ_CP049320.1 | Rep3_C | Caballeronia_sp._SBC2

Span: 1,803 bp | GC: 59.1% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)



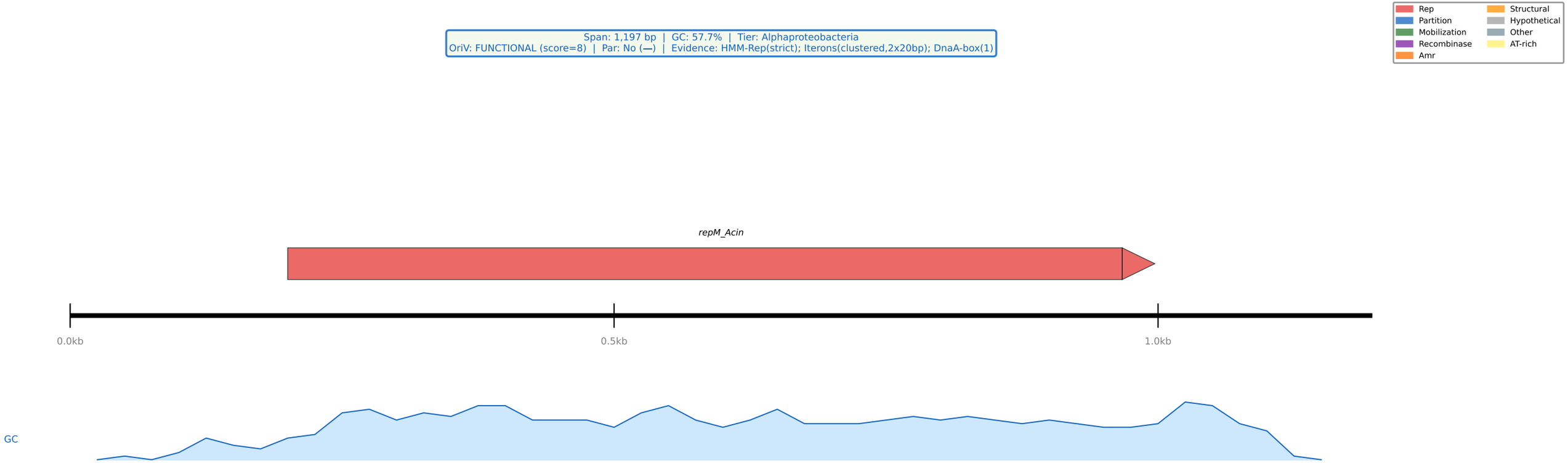
NZ_LC066381.1 | repM_Acin | Aureimonas_ureilytica

Span: 1,101 bp | GC: 59.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp); DnaA-box(1)



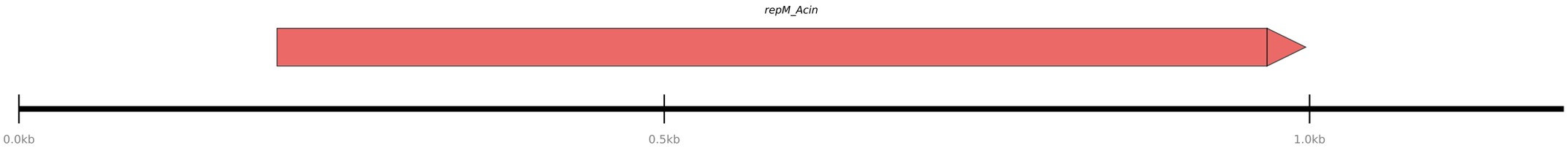
NZ_LC066394.1 | repM_Acin | Aureimonas sp._D3

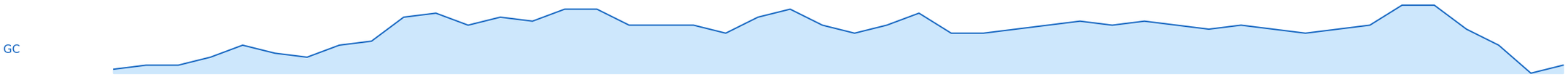
Span: 1,197 bp | GC: 57.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x20bp); DnaA-box(1)

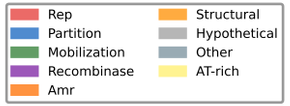


NZ_LC066398.1 | repM_Acin | Aureimonas sp._N4

Span: 1,197 bp | GC: 57.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=8) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x22bp); DnaA-box(1)

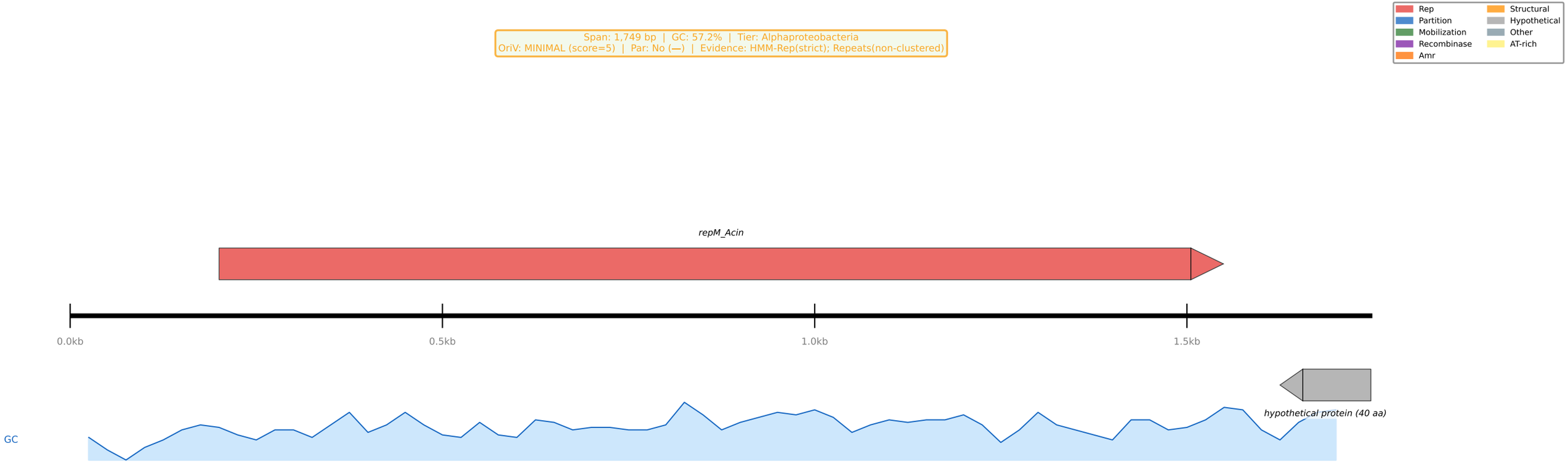






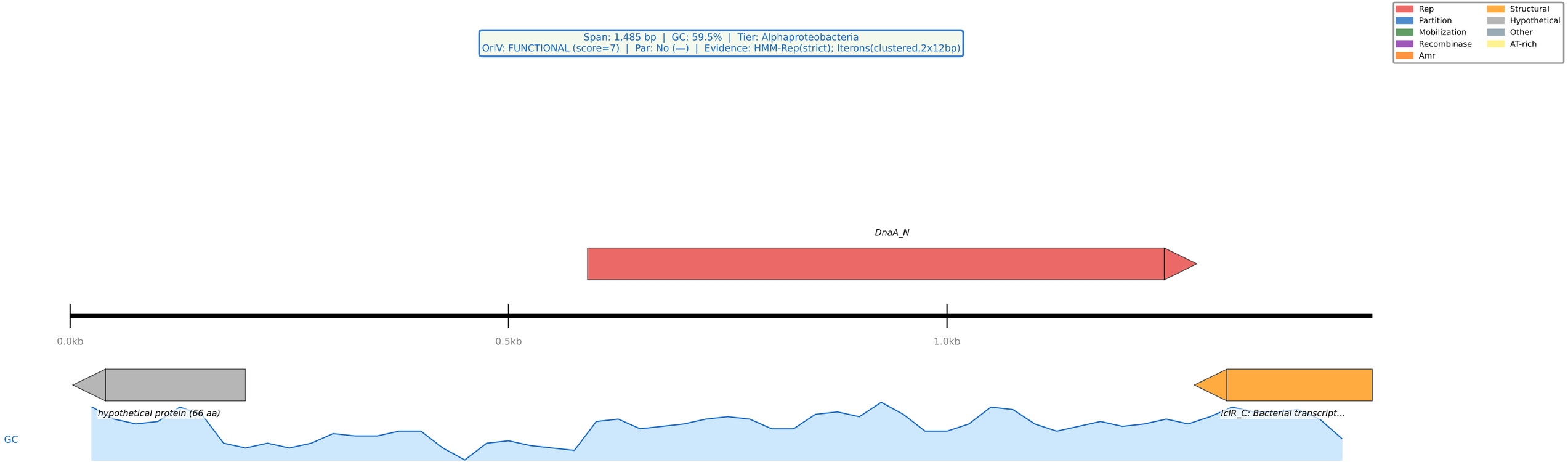
NZ_CP049829.1 | Rep3_C | Janthinobacterium_lividum

Span: 1,749 bp | GC: 57.2% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



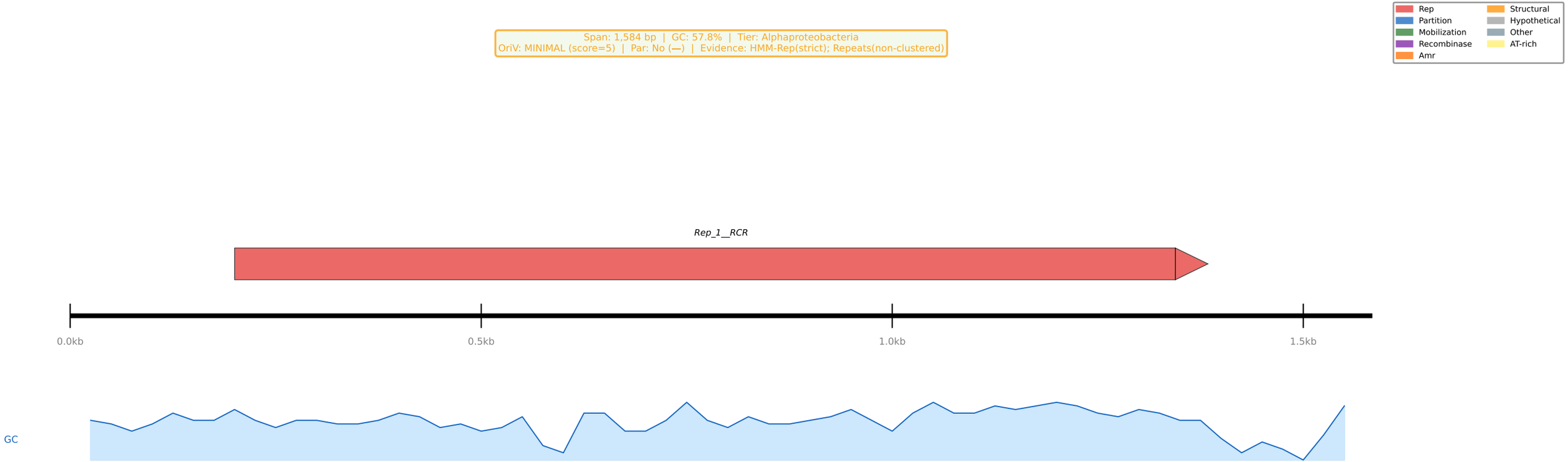
NZ_CP049346.1 | DnaA_N | Sulfitobacter_pontiacus

Span: 1,485 bp | GC: 59.5% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)



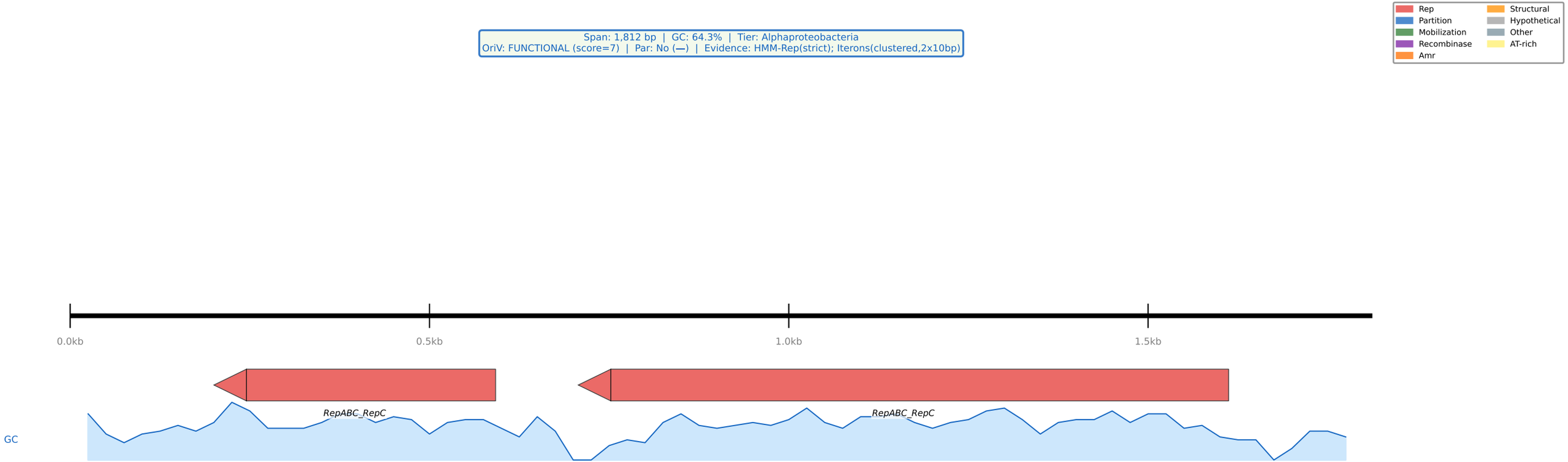
NZ_MG869624.1 | Rep_1__RCR | Polaromonas sp._W11N

Span: 1,584 bp | GC: 57.8% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



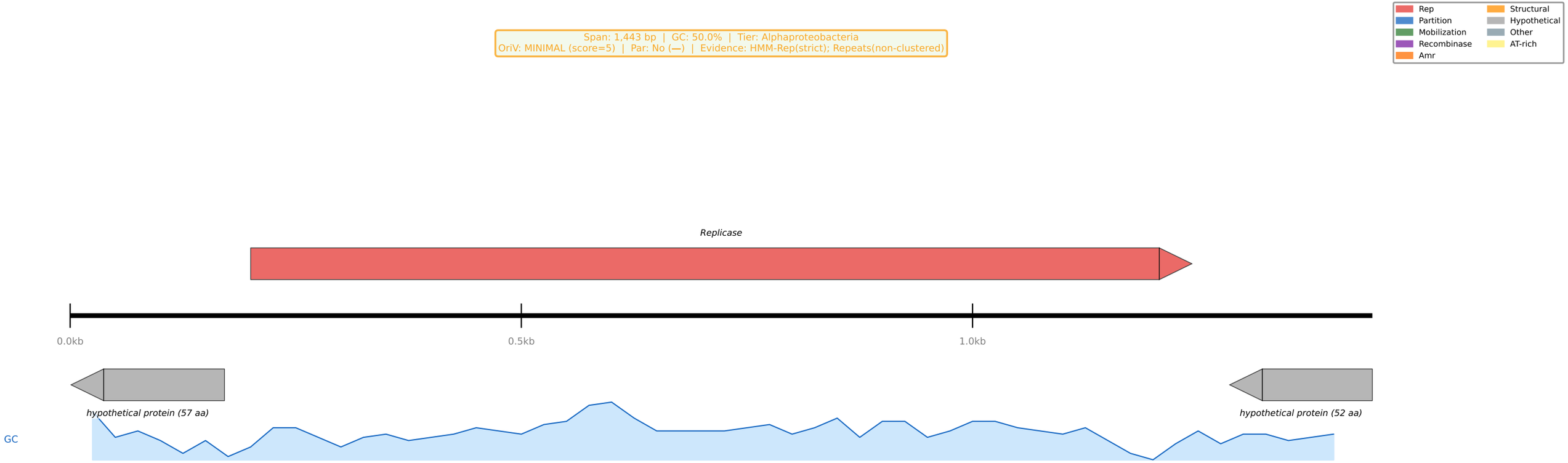
NZ_CP053924.1 | RepABC_RepC | Defluviicoccus_vanus

Span: 1,812 bp | GC: 64.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp)



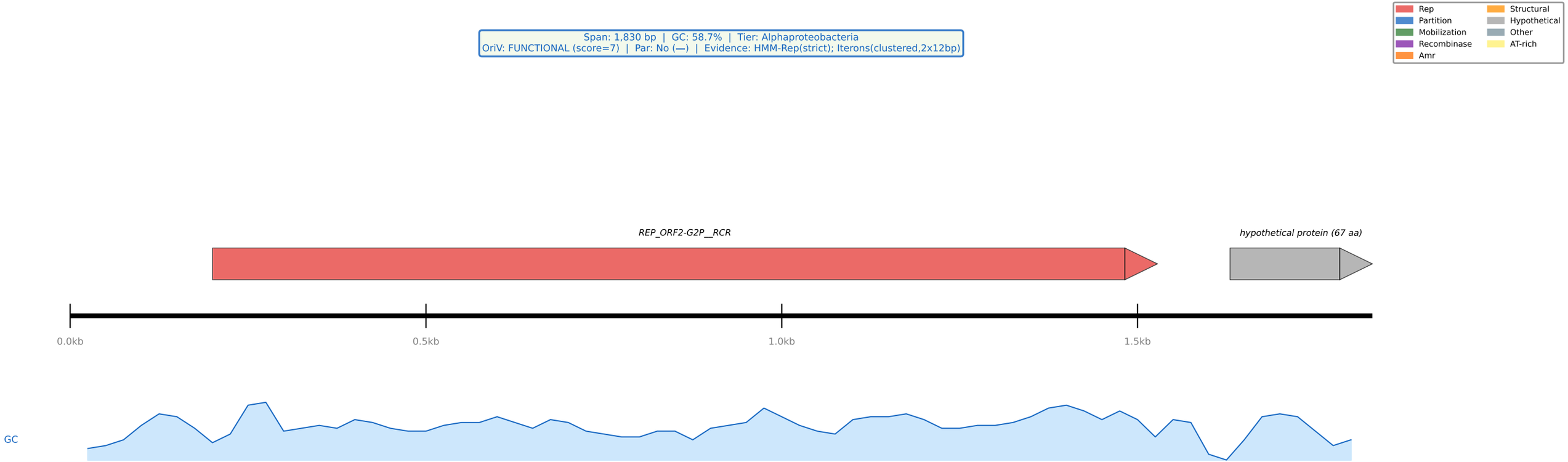
NZ_CP062963.1 | PriCT_1 | Rothia_terrae

Span: 1,443 bp | GC: 50.0% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



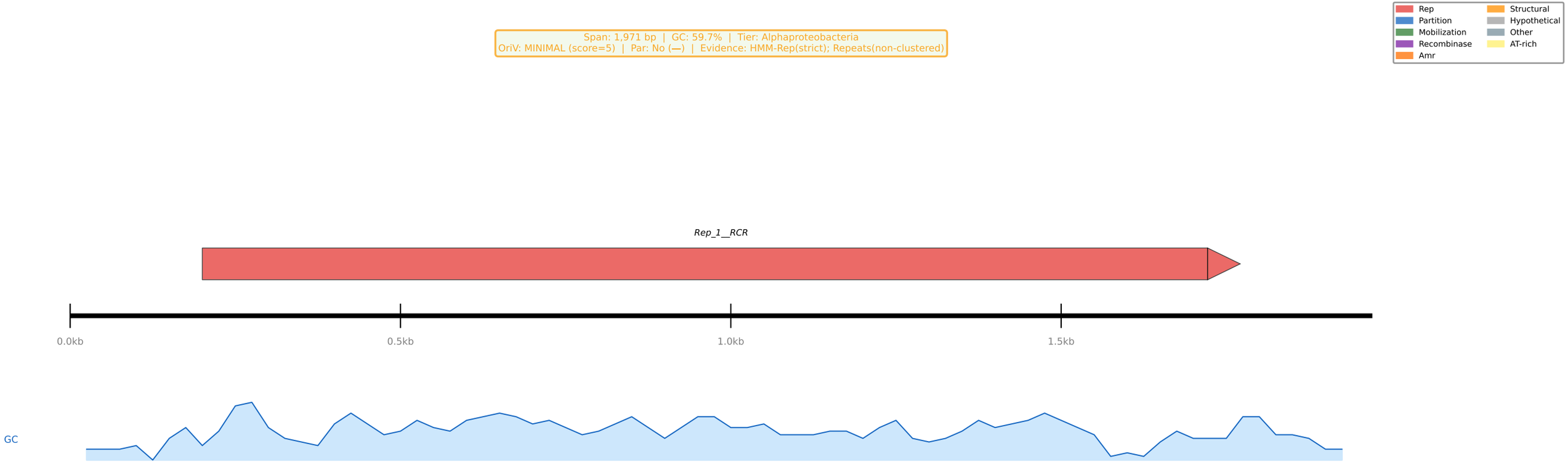
NZ_MK907532.1 | Rep_1__RCR | Aeromonas_sp.

Span: 1,830 bp | GC: 58.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)



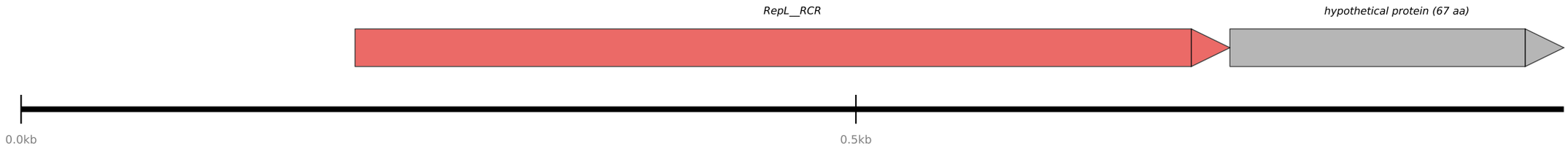
NZ_MK962686.1 | Rep_1__RCR | Aeromonas_sp.

Span: 1,971 bp | GC: 59.7% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



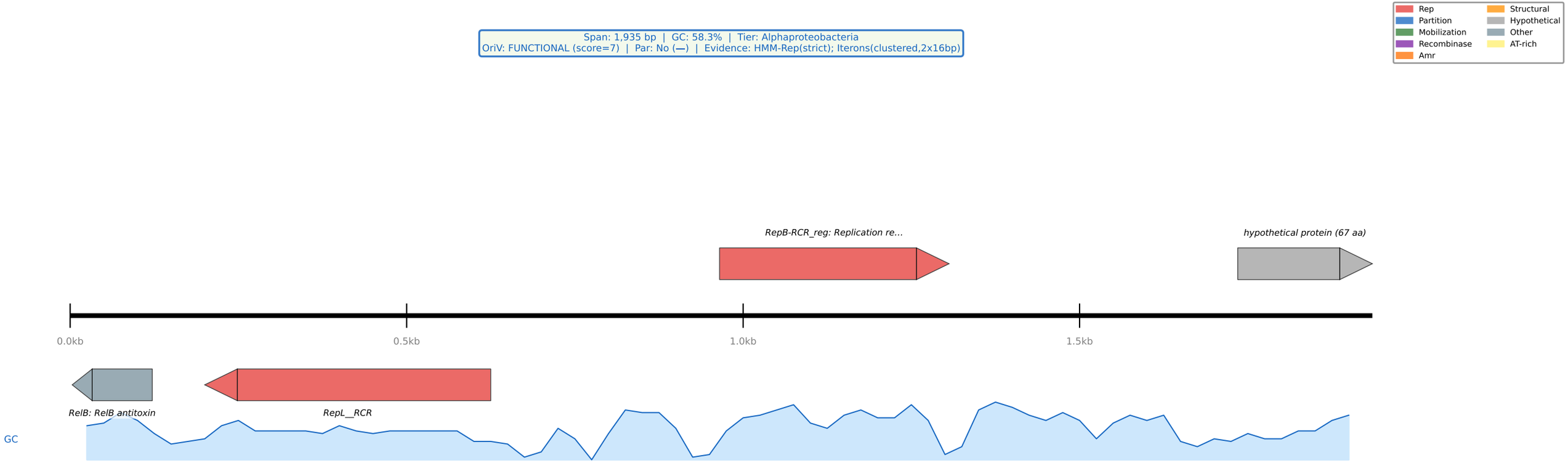
NZ_MK962688.1 | RepL__RCR | Aeromonas_sp.

Span: 924 bp | GC: 58.0% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x11bp)



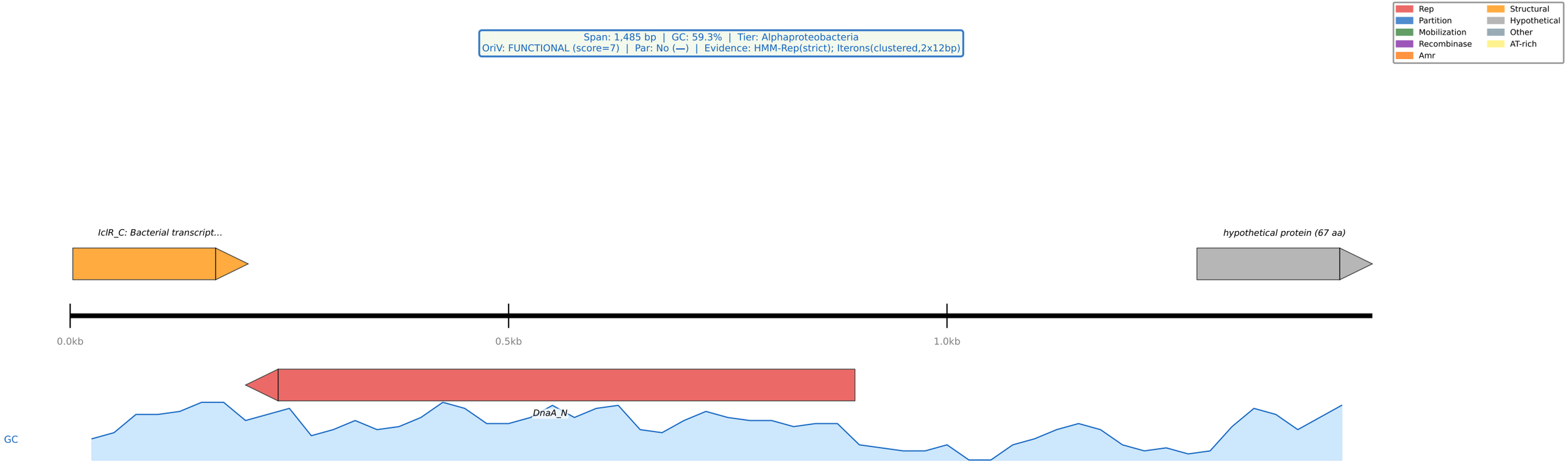
NZ_CP091178.1 | RepL__RCR | Aeromonas_caviae

Span: 1,935 bp | GC: 58.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x16bp)

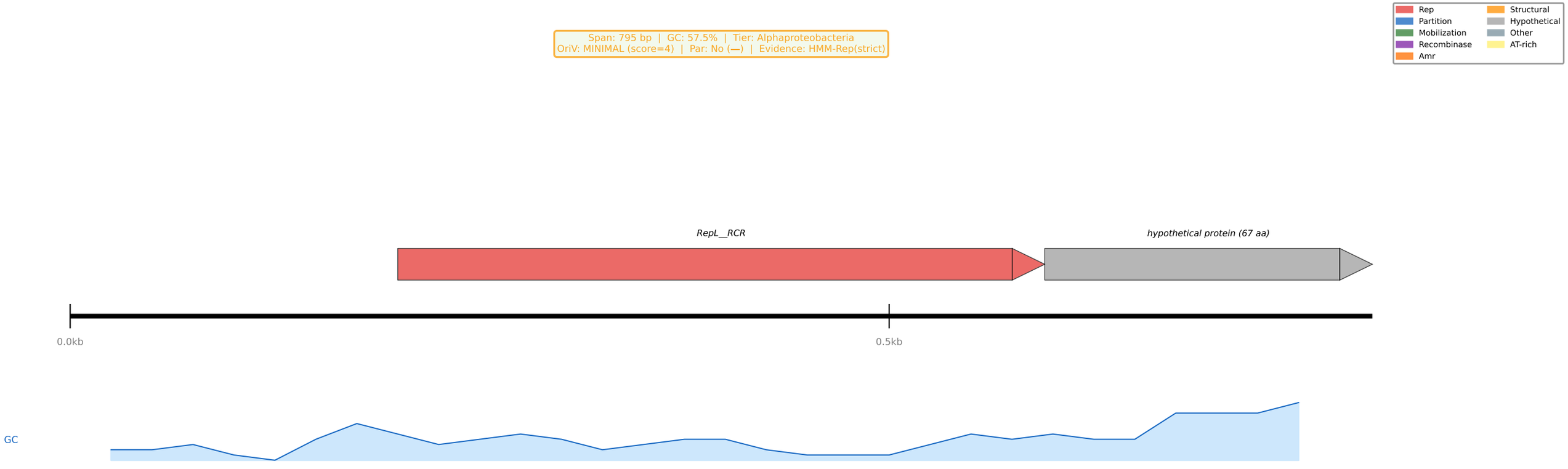


NZ_CP081118.1 | DnaA_N | Sulfitobacter_pontiacus

Span: 1,485 bp | GC: 59.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)

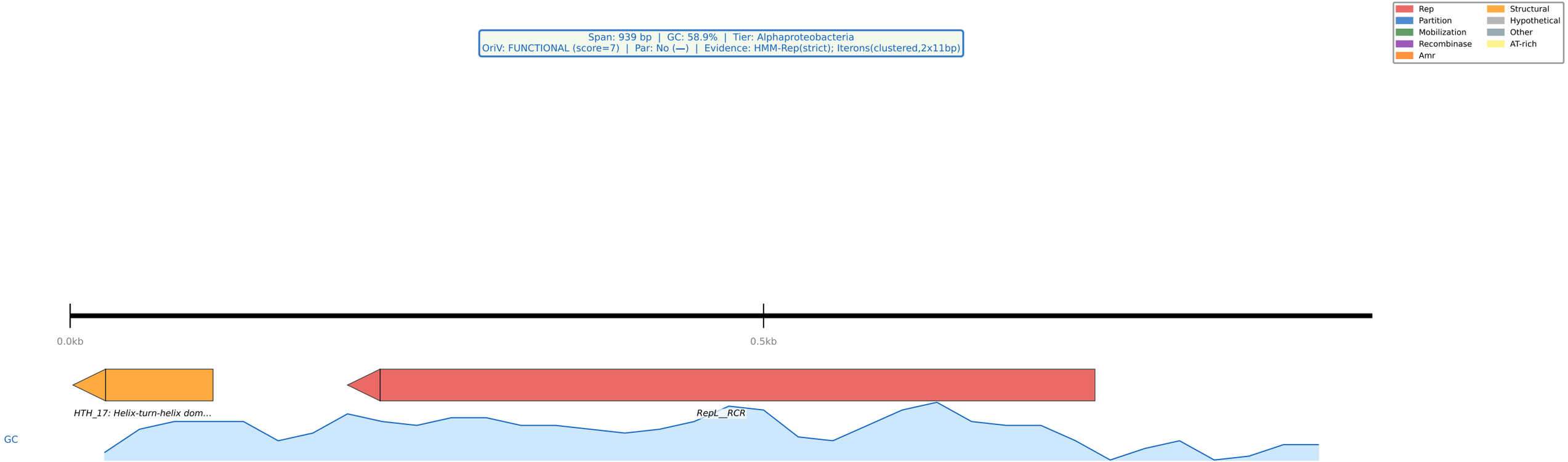


CP104739.1 | RepL__RCR | Edwardsiella_piscicida



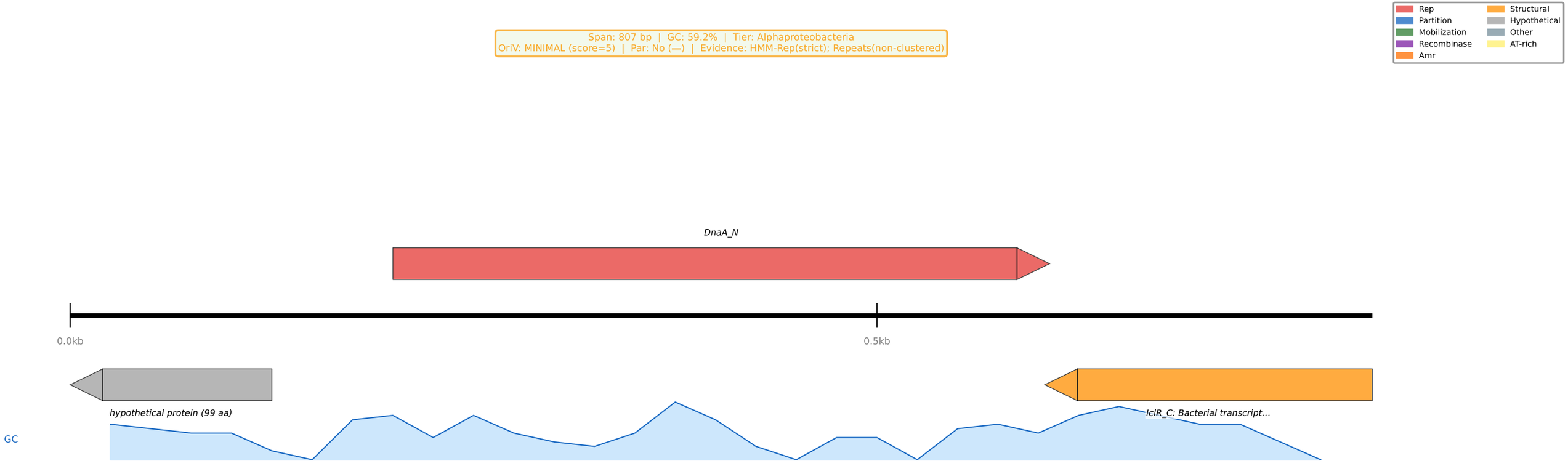
NZ_CP067139.1 | RepL__RCR | Paracoccus_fistulariae

Span: 939 bp | GC: 58.9% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x11bp)



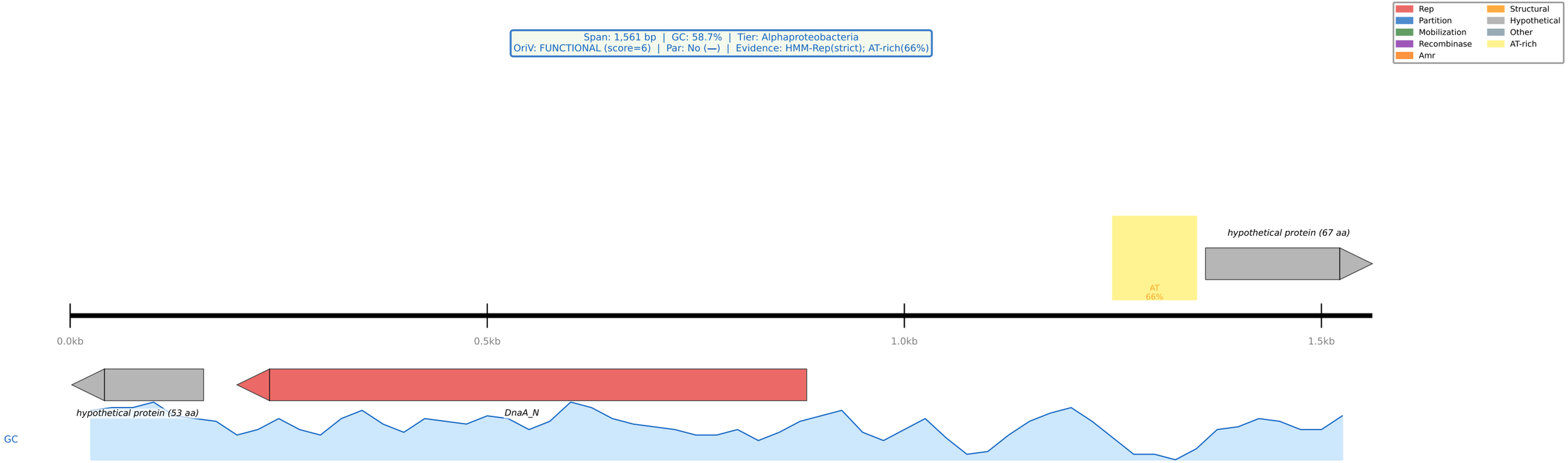
AP027374.1 | DnaA_N | Sulfitobacter_pontiacus

Span: 807 bp | GC: 59.2% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



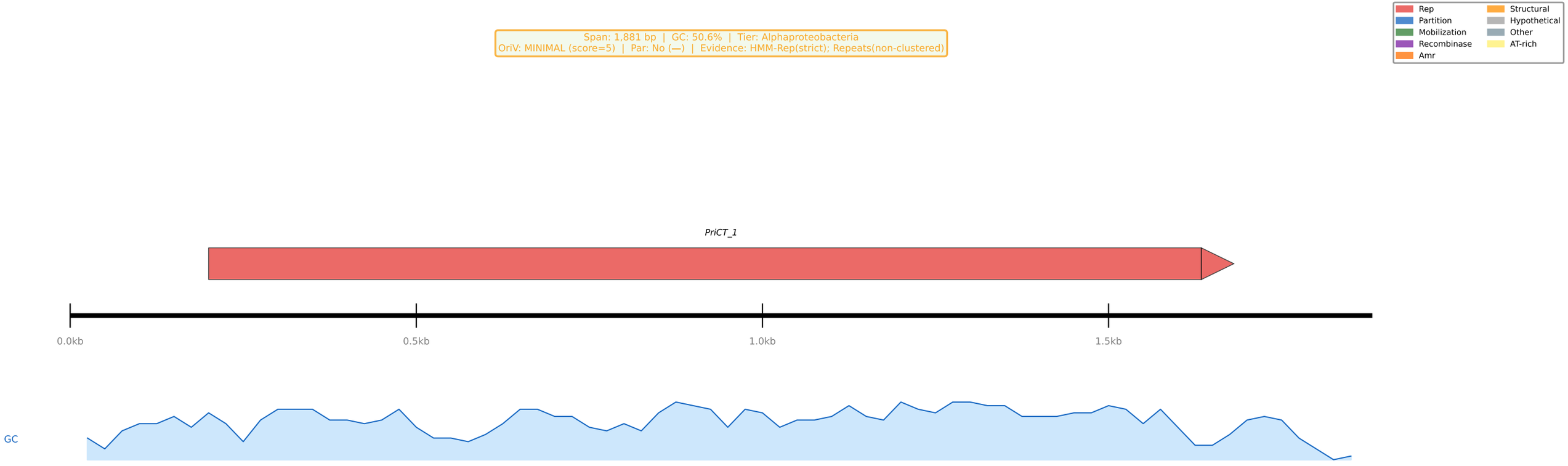
NZ_CP127166.1 | DnaA_N | Marivivens sp._LCG002

Span: 1,561 bp | GC: 58.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=6) | Par: No (—) | Evidence: HMM-Rep(strict); AT-rich(66%)



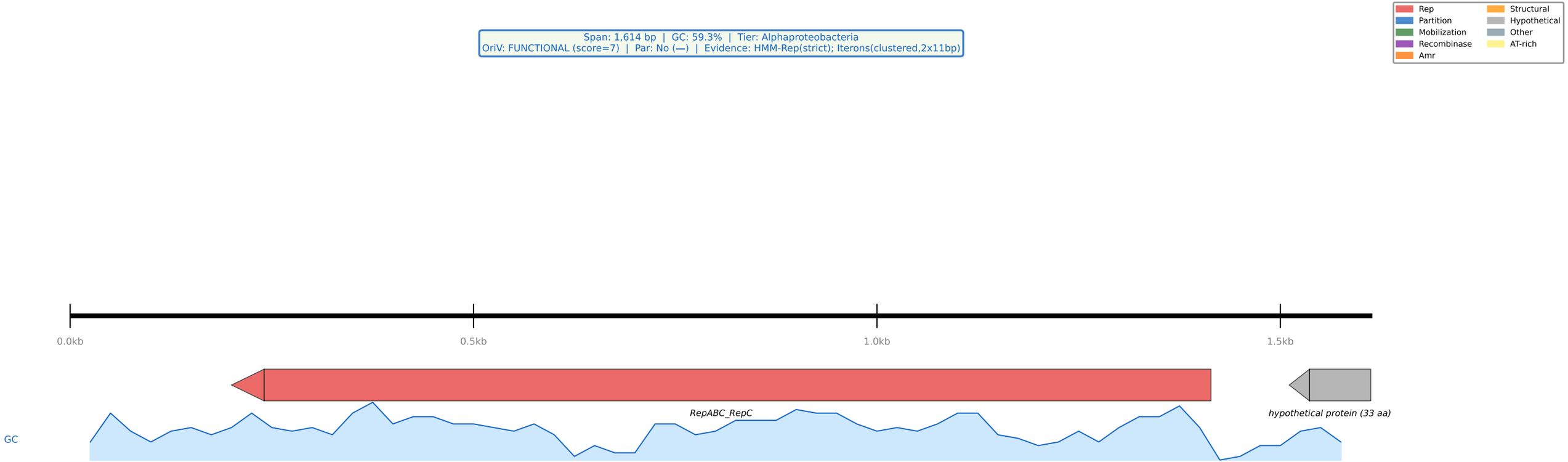
NZ_CP129659.1 | PriCT_1 | Lactiplantibacillus_pentosus

Span: 1,881 bp | GC: 50.6% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=5) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered)



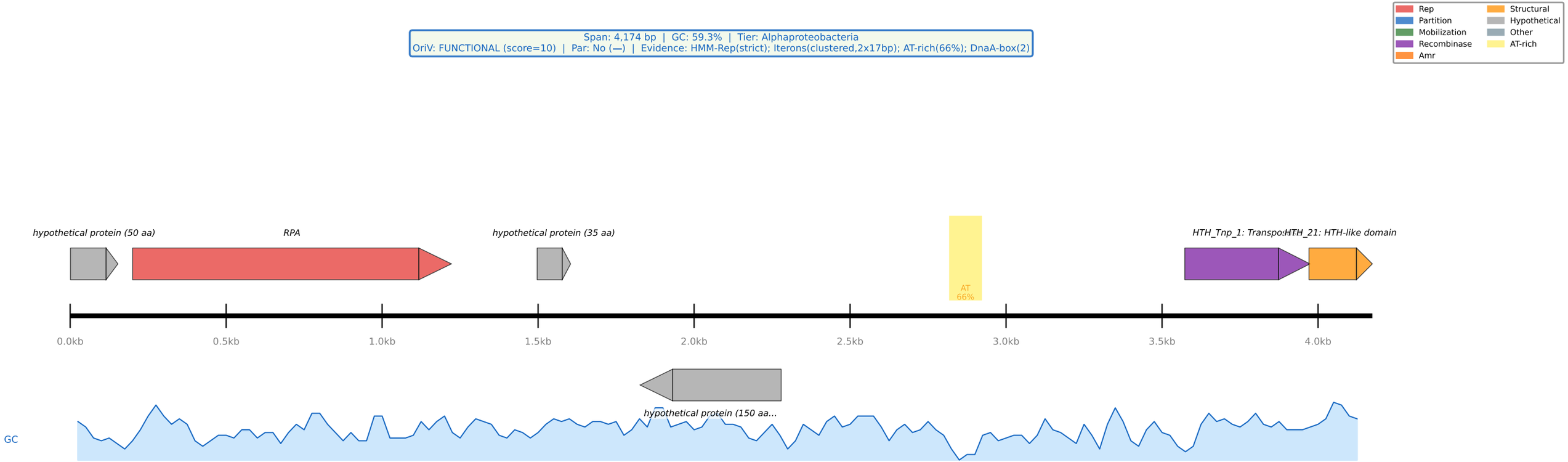
CP140962.1 | PRK13824.1 | Sinorhizobium_meliloti

Span: 1,614 bp | GC: 59.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x11bp)



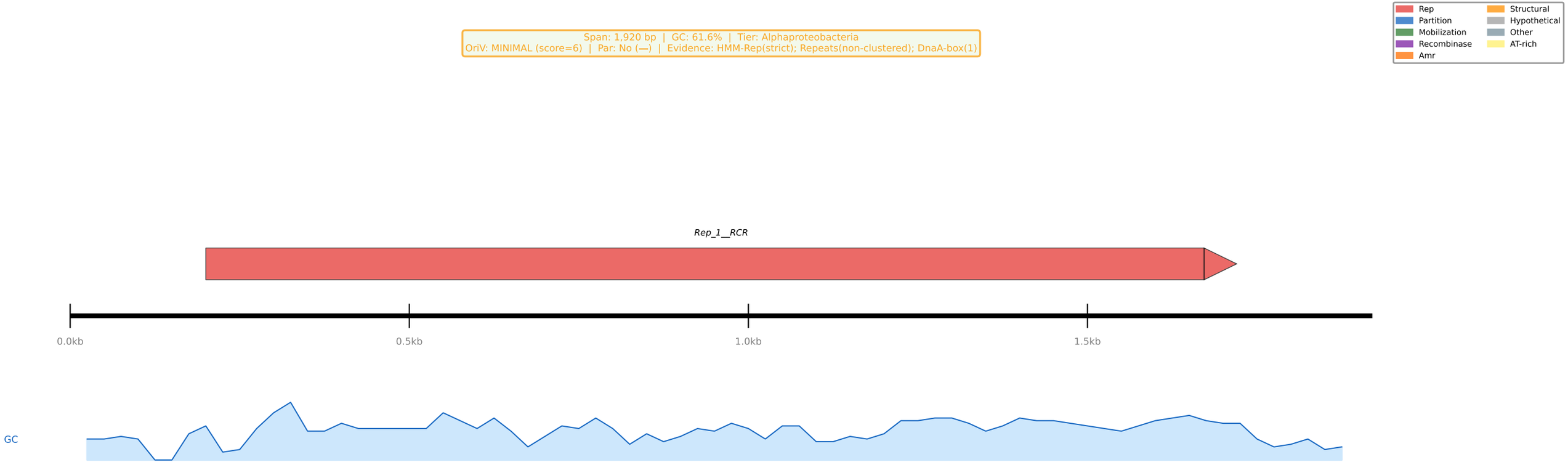
NZ_CP155076.1 | RPA | Paraburkholderia_sp._PREW-6R

Span: 4,174 bp | GC: 59.3% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=10) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x17bp); AT-rich(66%); DnaA-box(2)



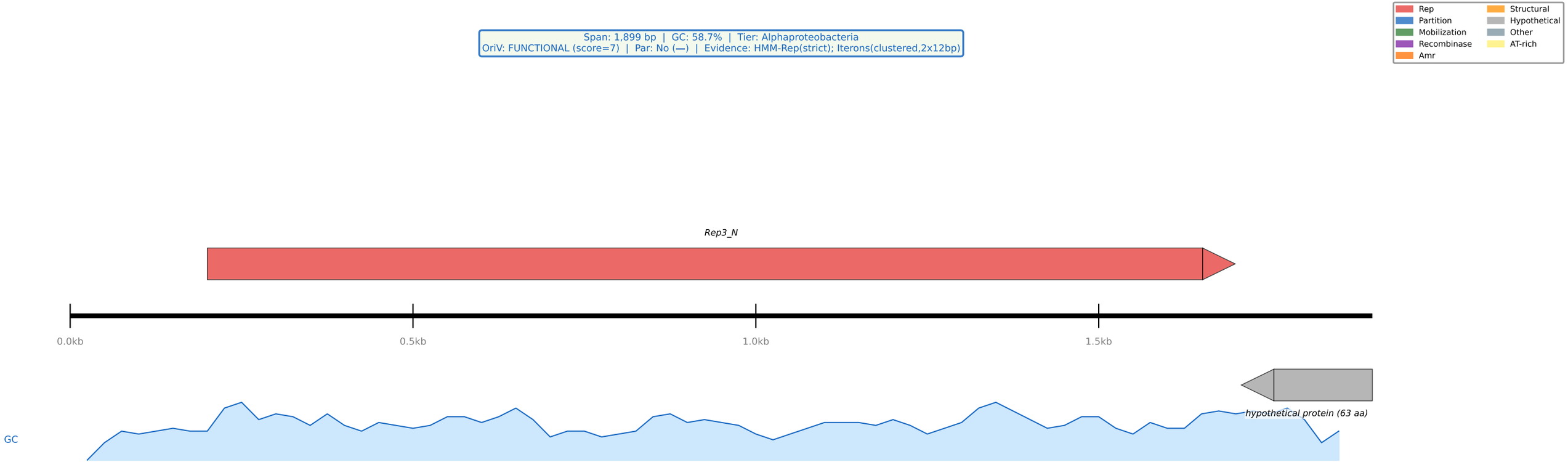
NC_004965.1 | Rep_1__RCR | Sinorhizobium_meliloti

Span: 1,920 bp | GC: 61.6% | Tier: Alphaproteobacteria
OriV: MINIMAL (score=6) | Par: No (—) | Evidence: HMM-Rep(strict); Repeats(non-clustered); DnaA-box(1)



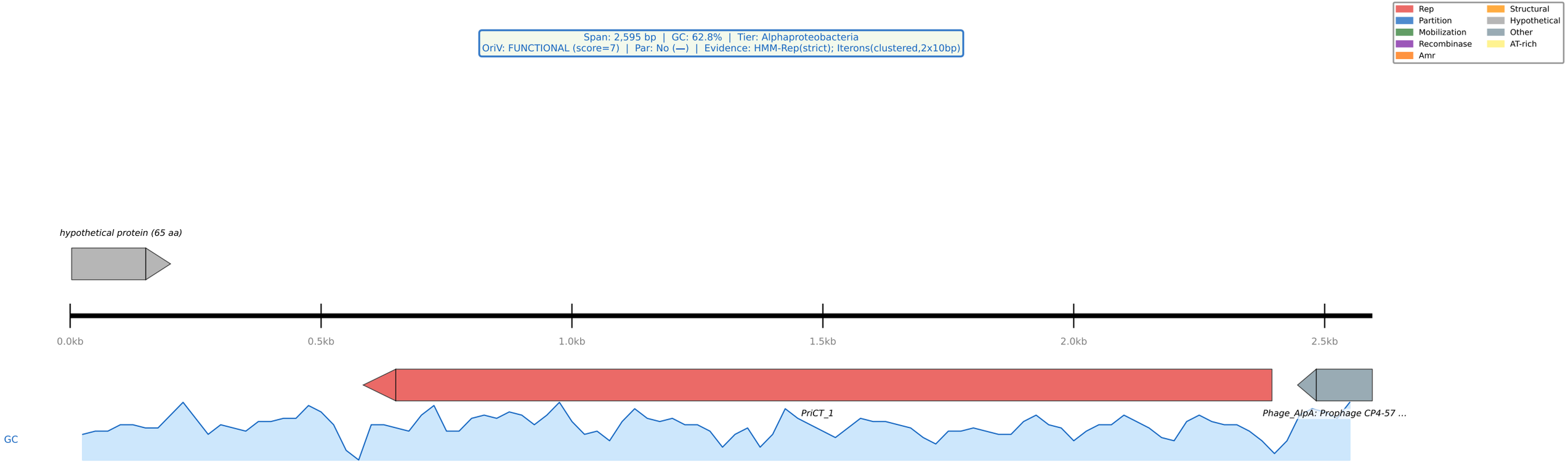
CP003091.1 | Rep3_N | Burkholderia_sp._YI23

Span: 1,899 bp | GC: 58.7% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x12bp)



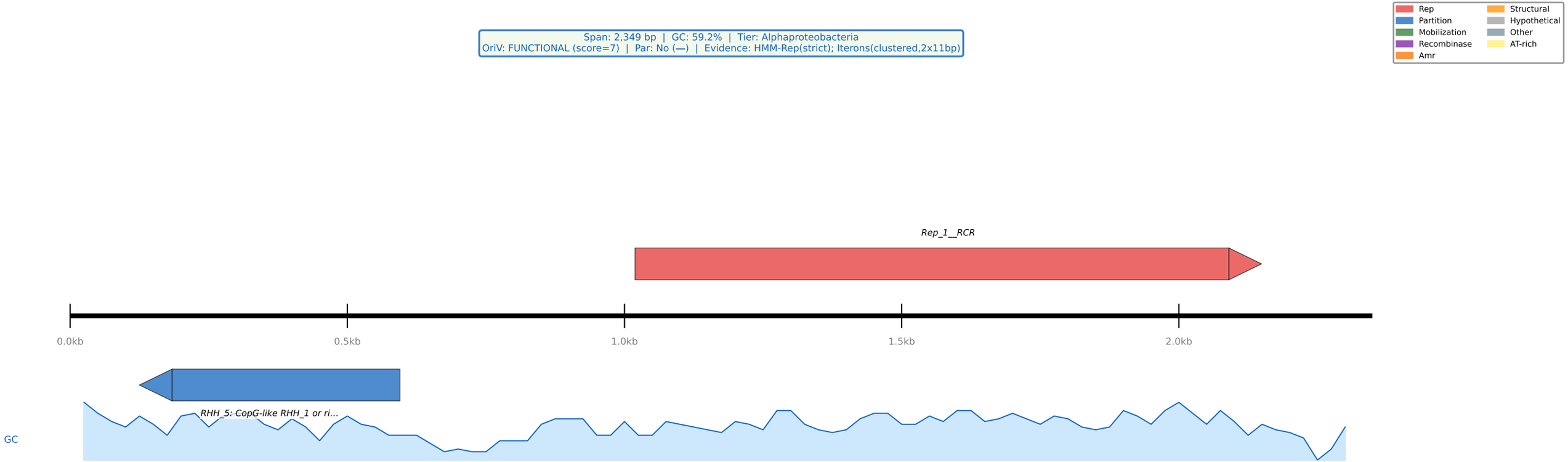
NC_006569.1 | PriCT_1 | Ruegeria_pomeroyi

Span: 2,595 bp | GC: 62.8% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp)



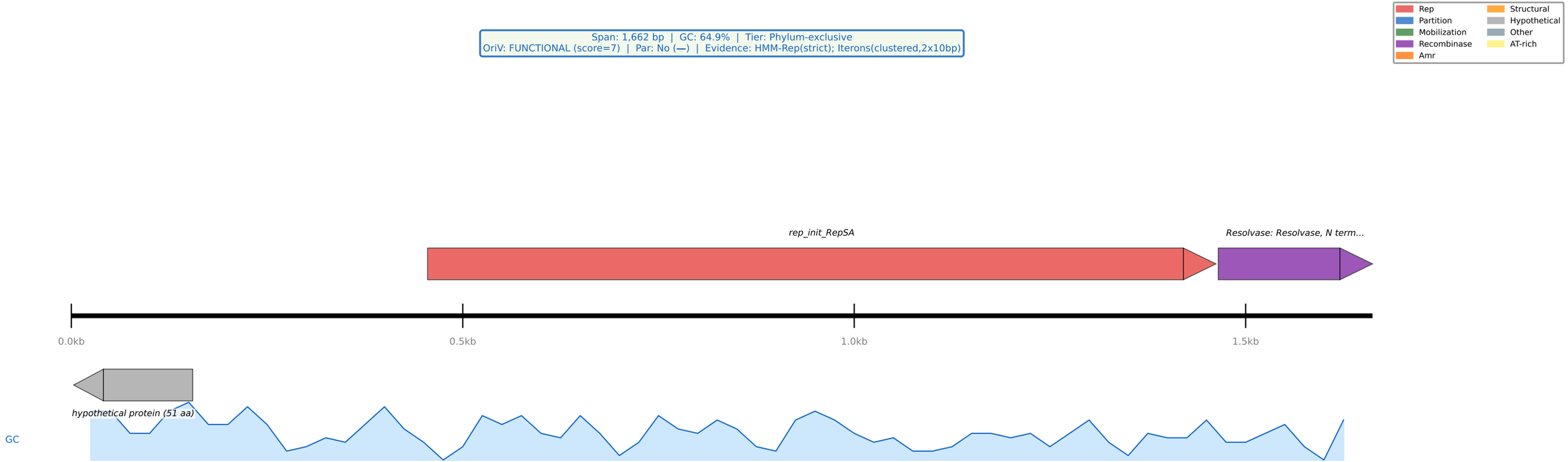
NC_007706.1 | Rep_1__RCR | Paramagnetospirillum_magneticum

Span: 2,349 bp | GC: 59.2% | Tier: Alphaproteobacteria
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x11bp)



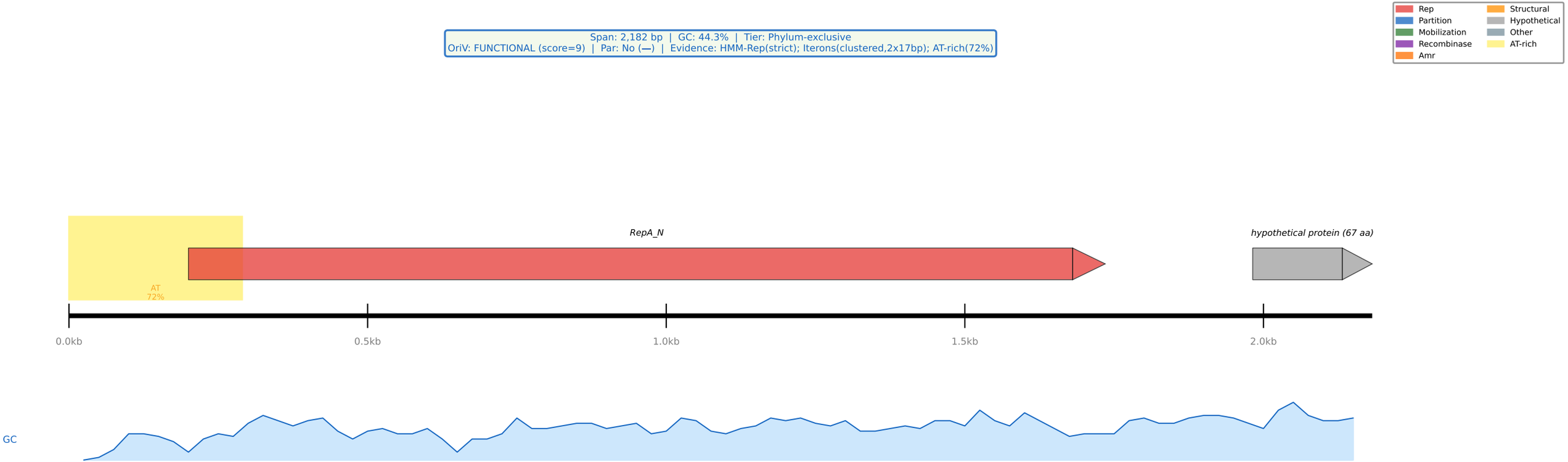
NZ_AP022578.1 | rep_init_RepSA | Mycolicibacterium_aubagnense

Span: 1,662 bp | GC: 64.9% | Tier: Phylum-exclusive
OriV: FUNCTIONAL (score=7) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp)



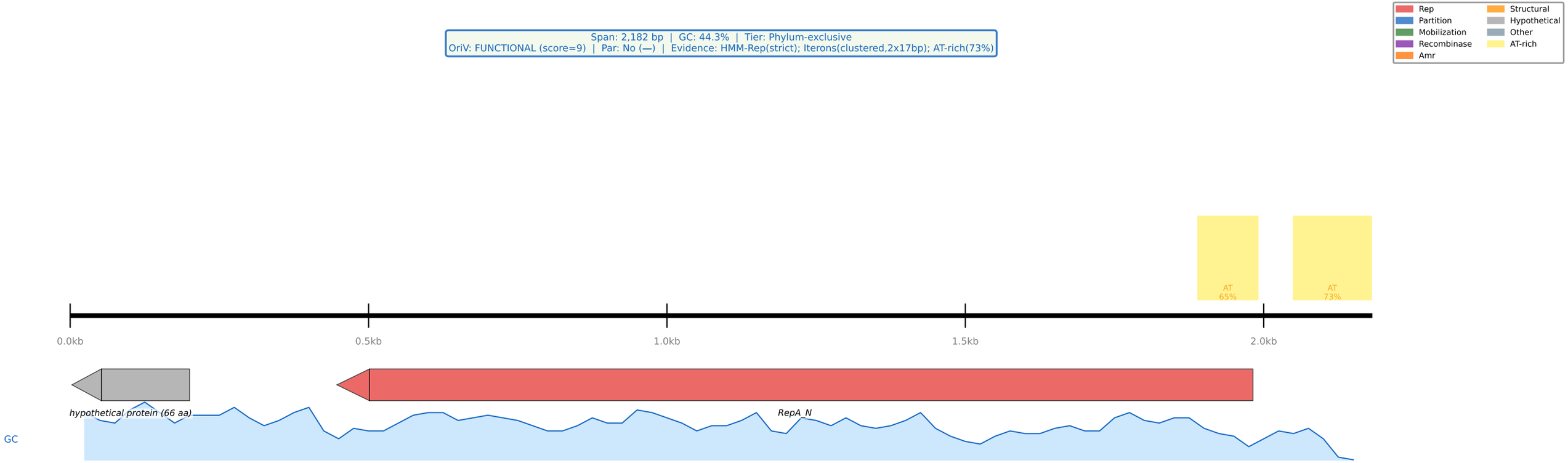
NZ_CP058955.1 | RepA_N | Limosilactobacillus_mucosae

Span: 2,182 bp | GC: 44.3% | Tier: Phylum-exclusive
OriV: FUNCTIONAL (score=9) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x17bp); AT-rich(72%)



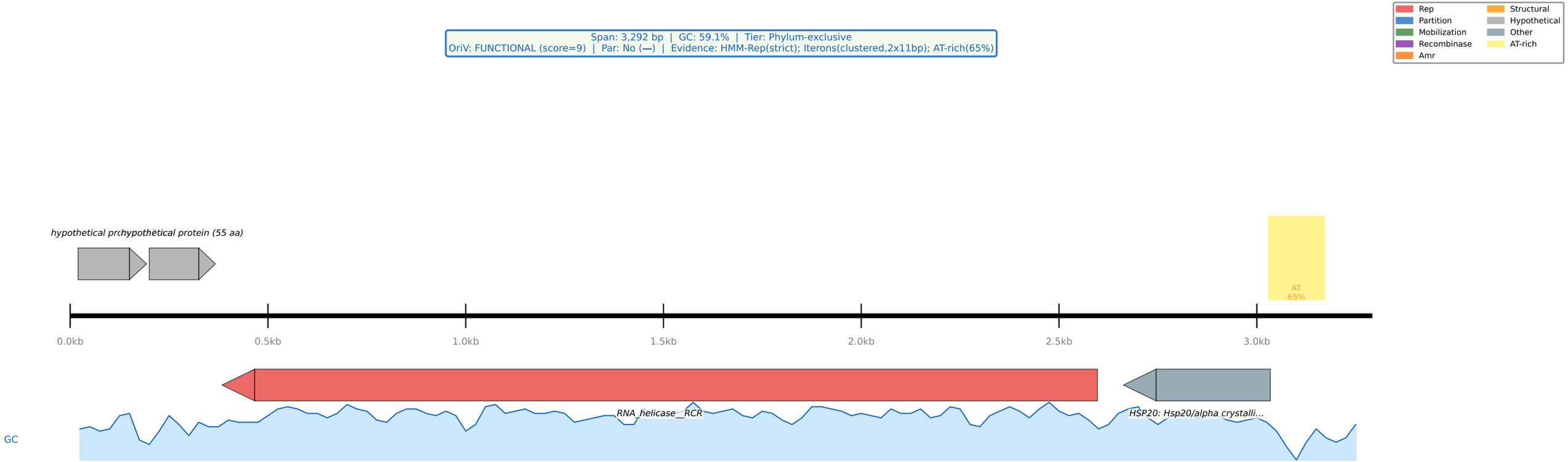
NZ_CP058956.1 | RepA_N | Limosilactobacillus_mucosae

Span: 2,182 bp | GC: 44.3% | Tier: Phylum-exclusive
OriV: FUNCTIONAL (score=9) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x17bp); AT-rich(73%)



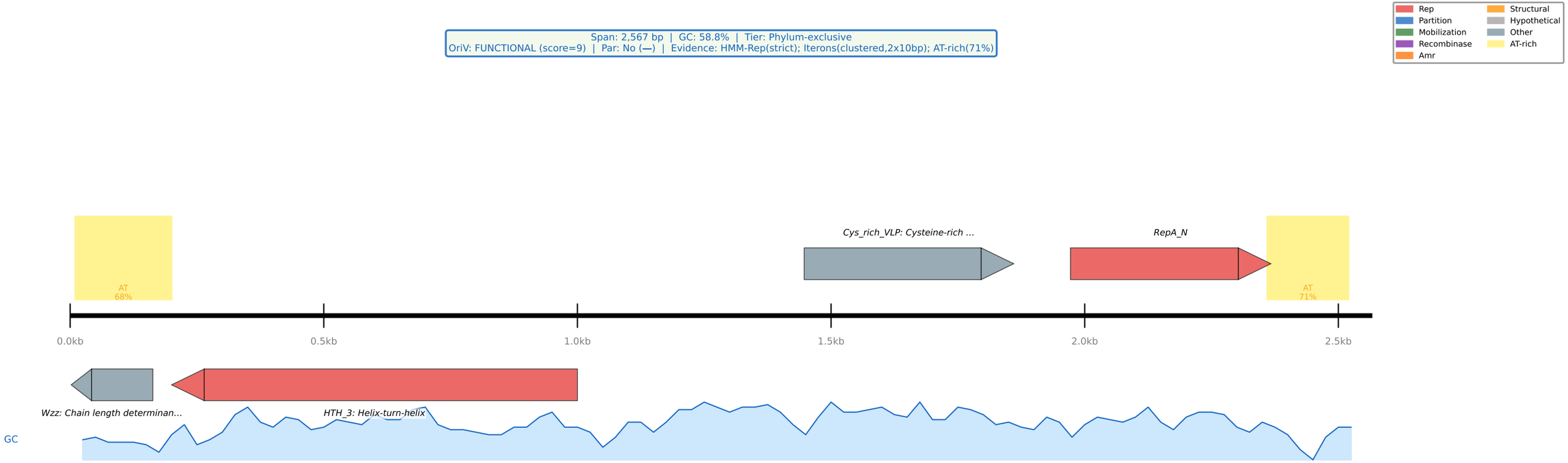
NZ_CP070335.2 | RNA_helicase__RCR | Halobacterium_sp._BOL4-2

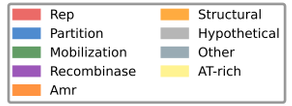
Span: 3,292 bp | GC: 59.1% | Tier: Phylum-exclusive
OriV: FUNCTIONAL (score=9) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x11bp); AT-rich(65%)



OY725877.1 | RepA_N | Subdoligranulum_variabile

Span: 2,567 bp | GC: 58.8% | Tier: Phylum-exclusive
OriV: FUNCTIONAL (score=9) | Par: No (—) | Evidence: HMM-Rep(strict); Iterons(clustered,2x10bp); AT-rich(71%)



OY725881.1 | RepA_N | Subdoligranulum_variabile

Span: 1,583 bp | GC: 54.9% | Tier: Phylum-exclusive
OriV: MINIMAL (score=4) | Par: No (—) | Evidence: HMM-Rep(strict)